

Attention and transformers

From one attention map to a trained transformer

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Section 1

Attention

Attention

Words can find their meaning in a sentence by **attending to** other words!

The baseball player swung his bat and hit a home run

Attention

Words can find their meaning in a sentence by **attending to** other words!



Attention

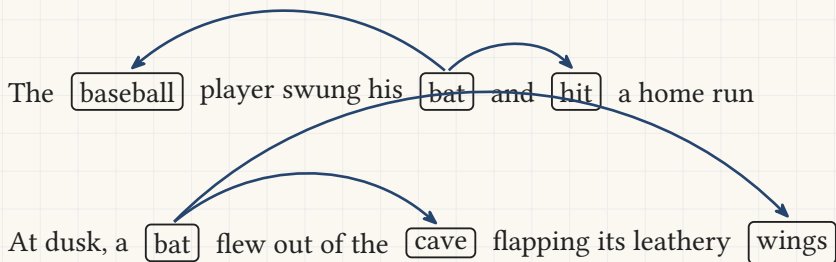
Words can find their meaning in a sentence by **attending to** other words!



At dusk, a bat flew out of the cave flapping its leathery wings

Attention

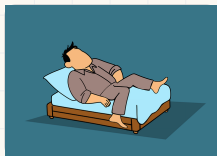
Words can find their meaning in a sentence by **attending to** other words!



Attention as information retrieval

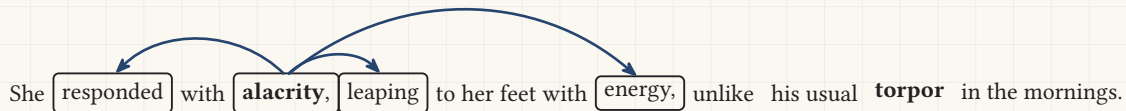
- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?

Attention as information retrieval



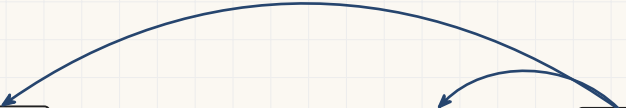
- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?
- Alacrity: brisk and cheerful readiness.
- Torpor: a state of physical or mental inactivity; lethargy.

Attention as information retrieval



Attention as information retrieval

She responded with **alacrity**, leaping to her feet with energy, **unlike** his usual **torpor** in the mornings.



Defining attention mechanisms

- We **define** a **value** representation for each word.

Defining attention mechanisms

- We **define** a **value** representation for each word.
- We let other words **enrich** their representations by **attending** to other words.

Creating embeddings for words

- Wuggish, a toy language: every sentence describes at most three things.

Creating embeddings for words

- Wuggish, a toy language: every sentence describes at most three things.



streept kat



bootet kat

Creating embeddings for words

- Wuggish, a toy language: every sentence describes at most three things.



streept kat



bootet kat



vennomus snaak wudden chaar



streept snaak streept kat wudden chaar

Creating embeddings for words

- Wuggish, a toy language: every sentence describes at most three things.



streept kat



bootet kat

Eight words

<i>kat</i>	cat
<i>snaak</i>	snake
<i>chaar</i>	chair
<i>streept</i>	striped
<i>bootet</i>	booted
<i>vennomus</i>	venomous
<i>wudden</i>	wooden
<i>brooken</i>	broken



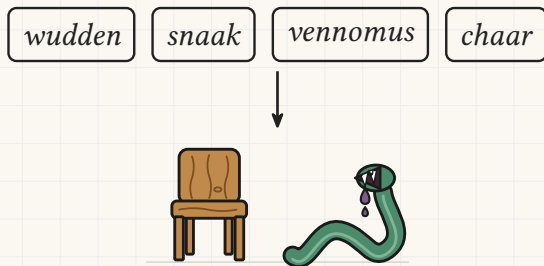
vennomus snaak wudden chaar



streept snaak streept kat wudden chaar

The Wuggish game

- Learn the eight words from pictures, then read bags of words in any order.



<https://eth-cscs-hs26.github.io/CSCS-PAGE/viz/part1/wuggish/?dev=0>

Also on the course website: Part I, Day 2, Attention and transformers.

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

Creating representations

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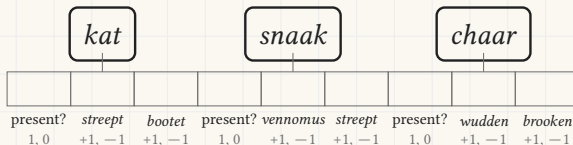
vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

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vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

	<i>kat</i>			<i>snaak</i>			<i>chaar</i>	
present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1

A

0	0	0	1	+1	+1	1	+1	-1
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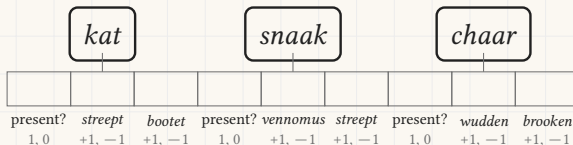
vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?



A	0	0	0	1	+1	+1	1	+1	-1
B	1	-1	-1	0	0	0	1	-1	+1

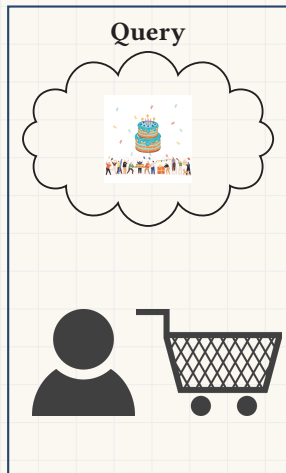


vennomus streept snaak wudden chaar

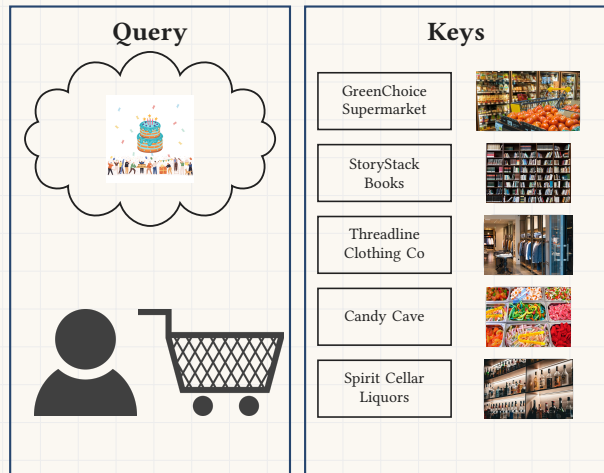


kat brooken chaar

Attention as information retrieval



Attention as information retrieval



Attention as information retrieval

Query



Keys

GreenChoice
Supermarket



StoryStack
Books



Threadline
Clothing Co



Candy Cave



Spirit Cellar
Liquors



Values



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Section 2

The inner product

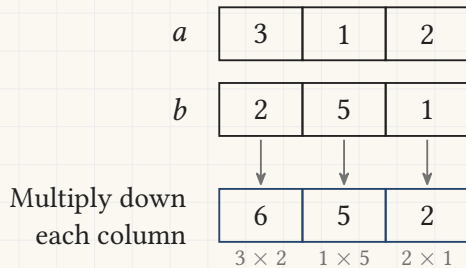
The inner product

$$a \quad \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline \end{array}$$

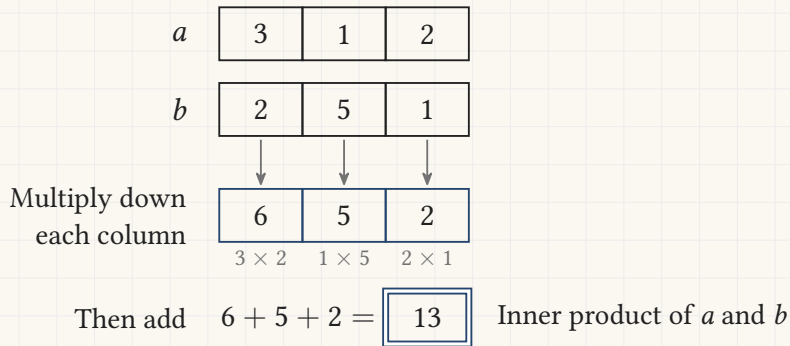
The inner product

a	3	1	2
b	2	5	1

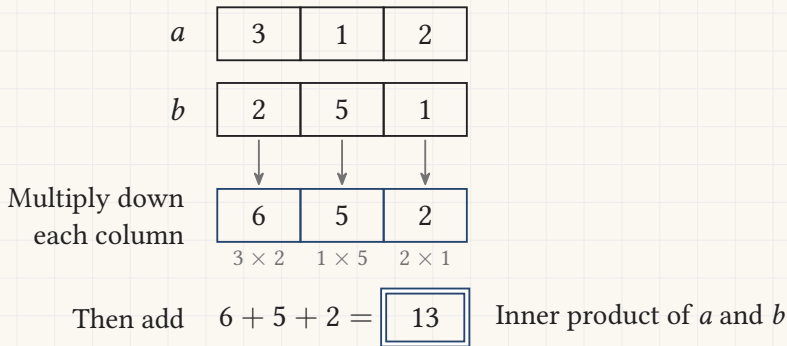
The inner product



The inner product



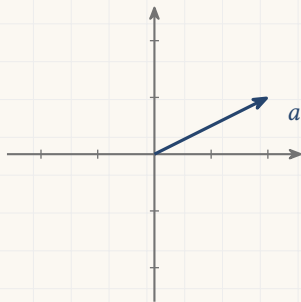
The inner product



$$\text{Inner product of } a \text{ and } b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

The inner product measures similarity

$$a = (2, 1)$$



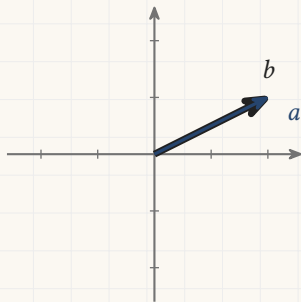
The inner product measures similarity

$$a = (2, 1)$$

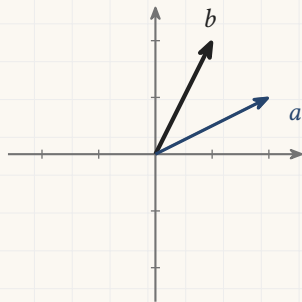
$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

Same direction



The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

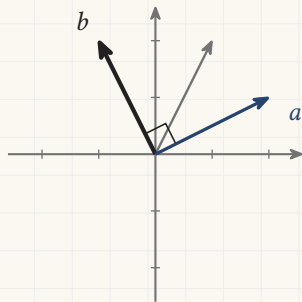
Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

Close directions

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

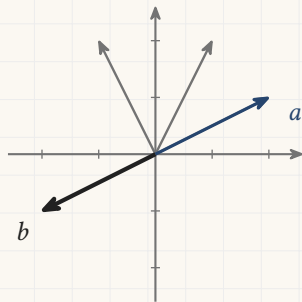
Close directions

$$b = (-1, 2)$$

$$2 \times (-1) + 1 \times 2 = 0$$

At right angles

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

Close directions

$$b = (-1, 2)$$

$$2 \times (-1) + 1 \times 2 = 0$$

At right angles

$$b = (-2, -1)$$

$$2 \times (-2) + 1 \times (-1) = -5$$

Opposite directions

The inner product of zeros and ones

Query of *kat*

<i>kat</i>	<i>snaak</i>	<i>chaar</i>
1	0	0

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1
Key of <i>wudden</i>	0	0	1	0

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1
Key of <i>wudden</i>	0	0	1	0
Key of <i>streept</i>	1	1	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	
Query of <i>kat</i>	1	0	0		
Key of <i>kat</i>	1	0	0	1	It counts the slots where both have a 1
Key of <i>bootet</i>	1	0	0	1	
Key of <i>wudden</i>	0	0	1	0	
Key of <i>streept</i>	1	1	0	1	

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Section 3

The attention mechanism

An attention map

wudden

kat

vennomus

chaar

bootet

snaak

brooken

An attention map

Queries

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
<i>wudden</i>	0	0	0
<i>kat</i>	1	0	0
<i>vennomus</i>	0	0	0
<i>chaar</i>	0	0	1
<i>bootet</i>	0	0	0
<i>snaak</i>	0	1	0
<i>brooken</i>	0	0	0

An attention map

	Queries			Keys		
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
<i>wudden</i>	0	0	0	0	0	1
<i>kat</i>	1	0	0	1	0	0
<i>vennomus</i>	0	0	0	0	1	0
<i>chaar</i>	0	0	1	0	0	1
<i>bootet</i>	0	0	0	1	0	0
<i>snaak</i>	0	1	0	0	1	0
<i>brooken</i>	0	0	0	0	0	1

An attention map

	Queries	Keys	Values																					
	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i>			<i>snaak</i>			<i>chaar</i>															
			present?	<i>streep</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streep</i>	present?	<i>wudden</i>	<i>brooken</i>													
<i>wudden</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>+2</td><td>0</td></tr></table>	0	+2	0				
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0	0	1																						
0	0	0																						
0	0	0																						
0	+2	0																						
<i>kat</i>	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0				
1	0	0																						
1	0	0																						
1	-1	-1																						
0	0	0																						
0	0	0																						
<i>vennomus</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>+2</td><td>0</td></tr></table>	0	+2	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0				
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0	1	0																						
0	0	0																						
0	+2	0																						
0	0	0																						
<i>chaar</i>	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1				
0	0	1																						
0	0	1																						
0	0	0																						
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1	-1	-1																						
<i>bootet</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	<table><tr><td>0</td><td>0</td><td>+2</td></tr></table>	0	0	+2	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0				
0	0	0																						
1	0	0																						
0	0	+2																						
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0	0	0																						
<i>snaak</i>	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0				
0	1	0																						
0	1	0																						
0	0	0																						
1	-1	-1																						
0	0	0																						
<i>brooken</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>+2</td></tr></table>	0	0	+2				
0	0	0																						
0	0	1																						
0	0	0																						
0	0	0																						
0	0	+2																						

An attention map

	Queries	Keys	Values			Embeddings		
	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> present? streept bootet	<i>snaak</i> present? vennomus streept	<i>chaar</i> present? wudden brooken	<i>kat</i> present? streept bootet	<i>snaak</i> present? vennomus streept	<i>chaar</i> present? wudden brooken
<i>wudden</i>	0 0 0	0 0 1	0 0 0	0 0 0	0 +2 0			
<i>kat</i>	1 0 0	1 0 0	1 -1 -1	0 0 0	0 0 0			
<i>vennomus</i>	0 0 0	0 1 0	0 0 0	0 +2 0	0 0 0			
<i>chaar</i>	0 0 1	0 0 1	0 0 0	0 0 0	1 -1 -1			
<i>bootet</i>	0 0 0	1 0 0	0 0 +2	0 0 0	0 0 0			
<i>snaak</i>	0 1 0	0 1 0	0 0 0	1 -1 -1	0 0 0			
<i>brooken</i>	0 0 0	0 0 1	0 0 0	0 0 0	0 0 +2			

An attention map

	Queries			Keys			Values									Embeddings								
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>			<i>snaak</i>			<i>chaar</i>			<i>kat</i>			<i>snaak</i>			<i>chaar</i>		
							<i>present?</i>	<i>streept</i>	<i>bootet</i>	<i>present?</i>	<i>vennomus</i>	<i>streept</i>	<i>present?</i>	<i>wudden</i>	<i>brooken</i>	<i>present?</i>	<i>streept</i>	<i>bootet</i>	<i>present?</i>	<i>vennomus</i>	<i>streept</i>	<i>present?</i>	<i>wudden</i>	<i>brooken</i>
<i>wudden</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	+2	0									
<i>kat</i>	1	0	0	1	0	0	1	-1	-1	0	0	0	0	0	0									
<i>vennomus</i>	0	0	0	0	1	0	0	0	0	0	+2	0	0	0	0									
<i>chaar</i>	0	0	1	0	0	1	0	0	0	0	0	0	1	-1	-1									
<i>bootet</i>	0	0	0	1	0	0	0	0	+2	0	0	0	0	0	0									
<i>snaak</i>	0	1	0	0	1	0	0	0	0	1	-1	-1	0	0	0									
<i>brooken</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	+2									

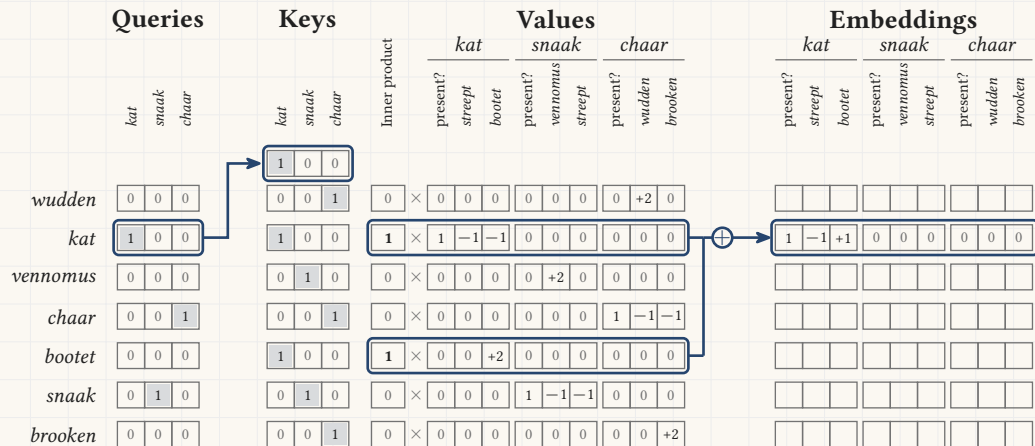
An attention map

	Queries			Keys				Values									Embeddings								
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	<i>kat</i>			<i>snaak</i>			<i>chaar</i>			<i>kat</i>			<i>snaak</i>			<i>chaar</i>		
								present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>	present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
<i>wudden</i>	0	0	0	1	0	0	0 × 0 0 0	0	0	0	0	0	0	0	+2	0									
<i>kat</i>	1	0	0	1	0	0	1 × 1 -1 -1	1	-1	-1	0	0	0	0	0	0									
<i>vennomus</i>	0	0	0	0	1	0	0 × 0 0 0	0	0	0	0	+2	0	0	0	0									
<i>chaar</i>	0	0	1	0	0	1	0 × 0 0 0	0	0	0	0	0	0	1	-1	-1									
<i>bootet</i>	0	0	0	1	0	0	1 × 0 0 +2	0	0	0	0	0	0	0	0	0									
<i>snaak</i>	0	1	0	0	1	0	0 × 0 0 0	0	0	0	1	-1	-1	0	0	0									
<i>brooken</i>	0	0	0	0	0	1	0 × 0 0 0	0	0	0	0	0	0	0	0	+2									

An attention map

Queries			Keys			Values			Embeddings		
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>		<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
									<i>present?</i>	<i>streept</i>	<i>bootet</i>
<i>wudden</i>	0	0	0		0	0	1	$0 \times 0 \ 0 \ 0$	0	0	0
<i>kat</i>	1	0	0		1	0	0	$1 \times 1 \ -1 \ -1$	0	0	0
<i>vennomus</i>	0	0	0		0	1	0	$0 \times 0 \ 0 \ 0$	0	+2	0
<i>chaar</i>	0	0	1		0	0	1	$0 \times 0 \ 0 \ 0$	0	0	0
<i>bootet</i>	0	0	0		1	0	0	$1 \times 0 \ 0 \ +2$	0	0	0
<i>snaak</i>	0	1	0		0	1	0	$0 \times 0 \ 0 \ 0$	1	-1	-1
<i>brooken</i>	0	0	0		0	0	1	$0 \times 0 \ 0 \ 0$	0	0	+2

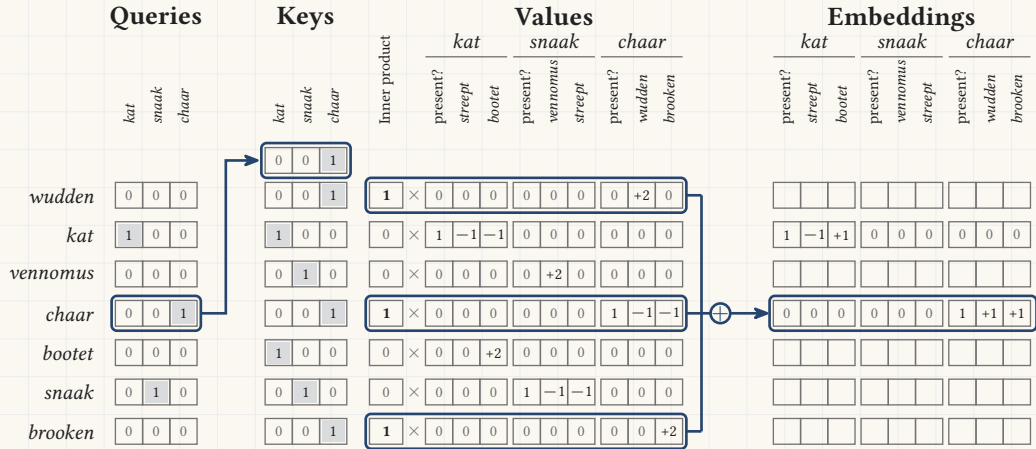
An attention map



An attention map

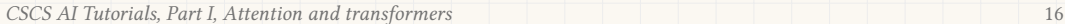
	Queries			Keys				Values									Embeddings															
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>present?</i>	<i>streept</i>	<i>bootet</i>	<i>present?</i>	<i>vennomus</i>	<i>streept</i>	<i>present?</i>	<i>wudden</i>	<i>brooken</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>present?</i>	<i>streept</i>	<i>bootet</i>	<i>present?</i>	<i>vennomus</i>	<i>streept</i>	<i>present?</i>	<i>wudden</i>	<i>brooken</i>	
<i>wudden</i>	0	0	0	0	0	1	1 ×	0	0	0	0	0	0	0	0	0	0	+2	0													
<i>kat</i>	1	0	0	1	0	0	0 ×	1	-1	-1	0	0	0	0	0	0	0	0	0	0	1	-1	+1	0	0	0	0	0	0	0	0	0
<i>vennomus</i>	0	0	0	0	1	0	0 ×	0	0	0	0	0	0	0	+2	0	0	0	0	0	0	0	0									
<i>chaar</i>	0	0	1	0	0	1	1 ×	0	0	0	0	0	0	0	0	0	1	-1	-1													
<i>bootet</i>	0	0	0	1	0	0	0 ×	0	0	+2	0	0	0	0	0	0	0	0	0													
<i>snaak</i>	0	1	0	0	1	0	0 ×	0	0	0	0	1	-1	-1	0	0	0	0	0													
<i>brooken</i>	0	0	0	0	0	1	1 ×	0	0	0	0	0	0	0	0	0	0	0	+2													

An attention map



An attention map

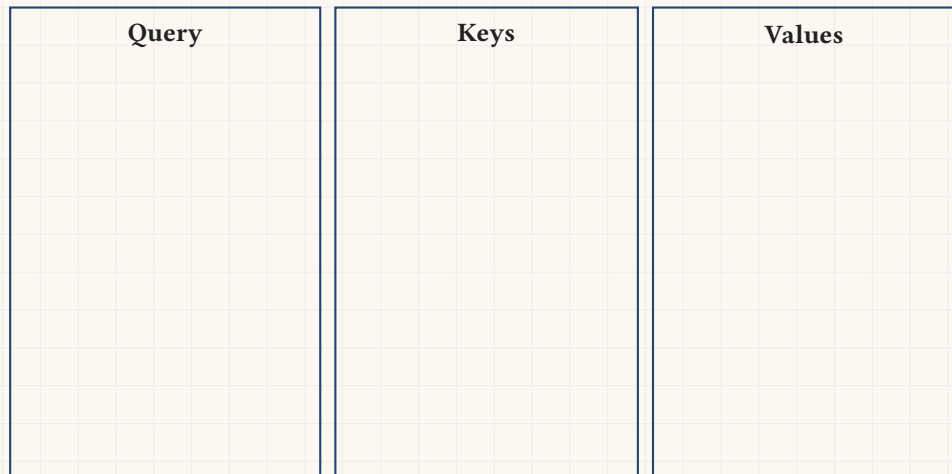
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<i>wudden</i>	0	0	0	0	0	1	0 ×	0	0	0	0	0	0	0	0	0	0	+2	0													
<i>kat</i>	1	0	0	1	0	0	0 ×	1	-1	-1	0	0	0	0	0	0	0	0	0	0	1	-1	+1	0	0	0	0	0	0	0	0	0
<i>vennomus</i>	0	0	0	0	1	0	1 ×	0	0	0	0	0	0	0	+2	0	0	0	0	0	0	0	0									
<i>chaar</i>	0	0	1	0	0	1	0 ×	0	0	0	0	0	0	0	0	0	1	-1	-1	0	0	0	0	0	0	0	1	+1	+1			
<i>bootet</i>	0	0	0	1	0	0	0 ×	0	0	+2	0	0	0	0	0	0	0	0	0													
<i>snaak</i>	0	1	0	0	1	0	1 ×	0	0	0	1	-1	-1	0	0	0	0	0	0													
<i>brooken</i>	0	0	0	0	0	1	0 ×	0	0	0	0	0	0	0	0	0	0	0	+2													



An attention map

	Queries	Keys	Values			Embeddings																										
	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> present? <i>streept</i> <i>bootet</i>	<i>snaak</i> present? <i>vennomus</i> <i>streept</i>	<i>chaar</i> present? <i>wudden</i> <i>brooken</i>	<i>kat</i> present? <i>streept</i> <i>bootet</i>	<i>snaak</i> present? <i>vennomus</i> <i>streept</i>	<i>chaar</i> present? <i>wudden</i> <i>brooken</i>																								
<i>wudden</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>+2</td><td>0</td></tr></table>	0	+2	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0
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<i>vennomus</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>+2</td><td>0</td></tr></table>	0	+2	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0
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<i>chaar</i>	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>+1</td><td>+1</td></tr></table>	1	+1	+1
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<i>bootet</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	<table><tr><td>0</td><td>0</td><td>+2</td></tr></table>	0	0	+2	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0
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<i>snaak</i>	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>+1</td><td>-1</td></tr></table>	1	+1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0
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<i>brooken</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>+2</td></tr></table>	0	0	+2	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0
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An attention map for location of objects



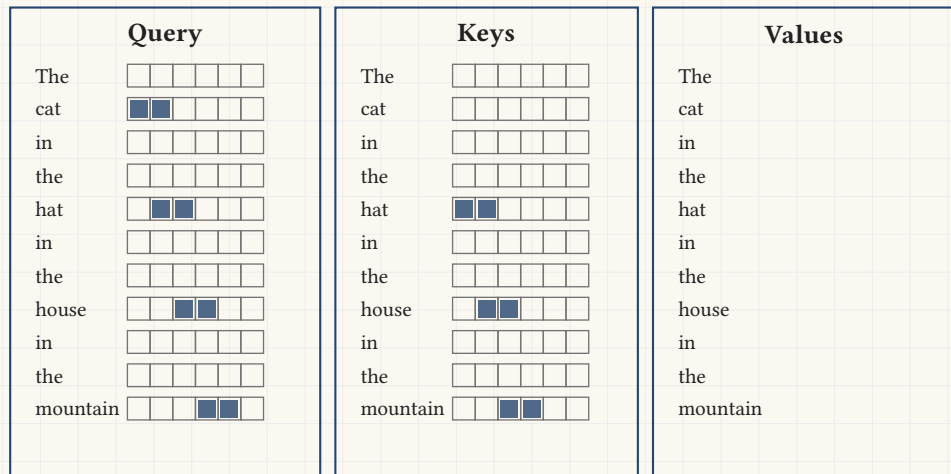
An attention map for location of objects

Query	Keys	Values
The cat in the hat in the house in the mountain	The cat in the hat in the house in the mountain	The cat in the hat in the house in the mountain

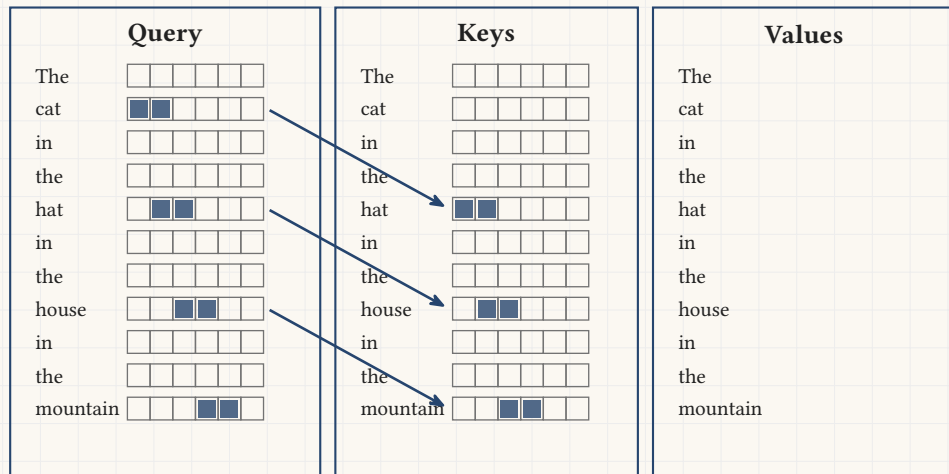
An attention map for location of objects

Query		Keys	Values
The		The	The
cat		cat	cat
in		in	in
the		the	the
hat		hat	hat
in		in	in
the		the	the
house		house	house
in		in	in
the		the	the
mountain		mountain	mountain

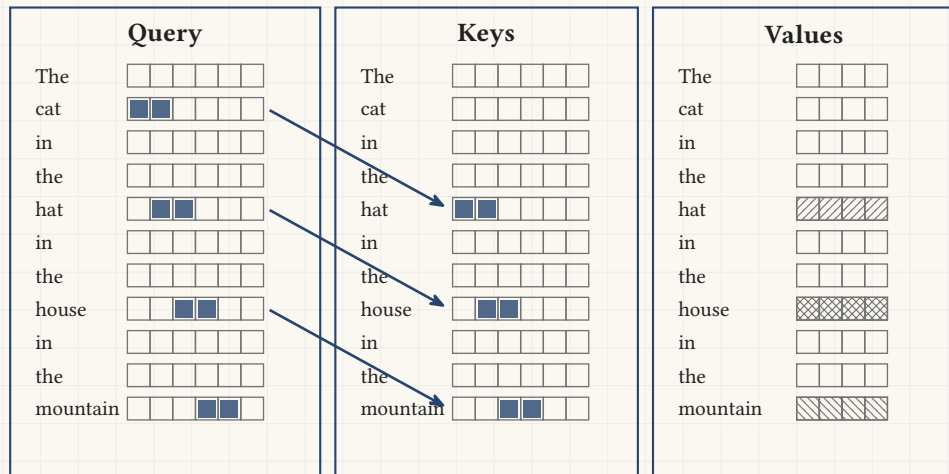
An attention map for location of objects



An attention map for location of objects



An attention map for location of objects

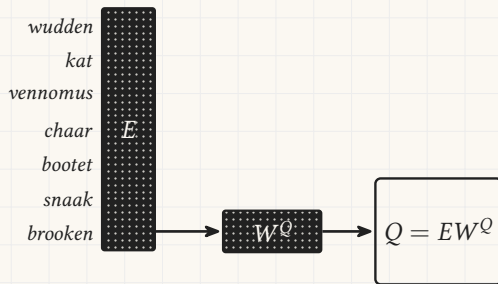


Abstraction of an attention mechanism

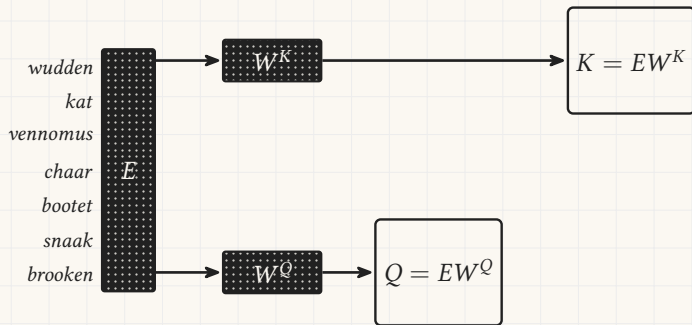
wudden
kat
vennomus
chaar *E*
bootet
snaak
brooken



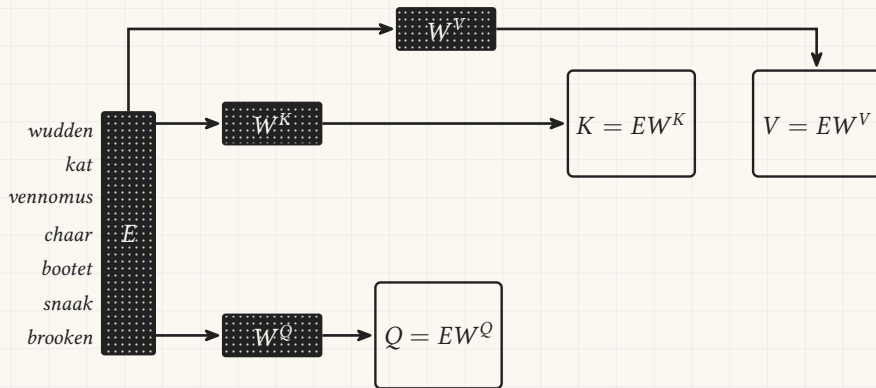
Abstraction of an attention mechanism



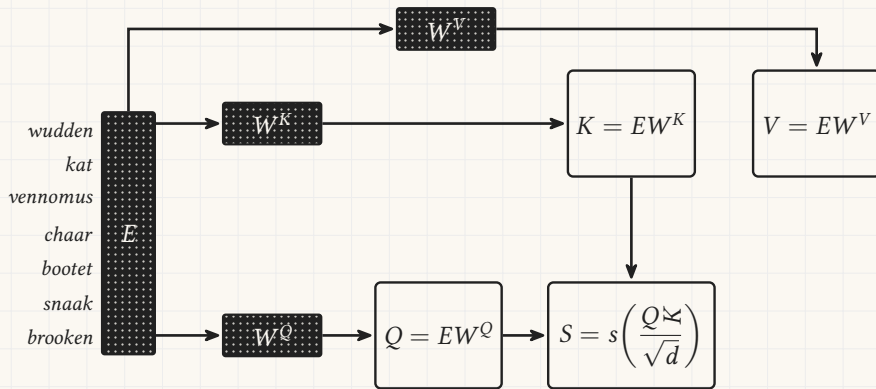
Abstraction of an attention mechanism



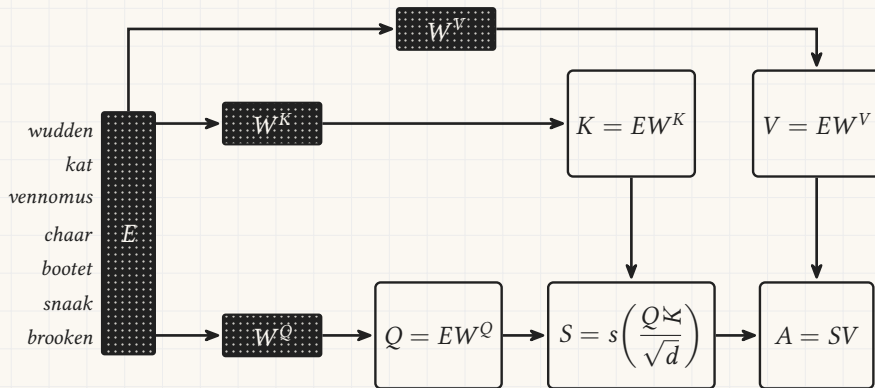
Abstraction of an attention mechanism



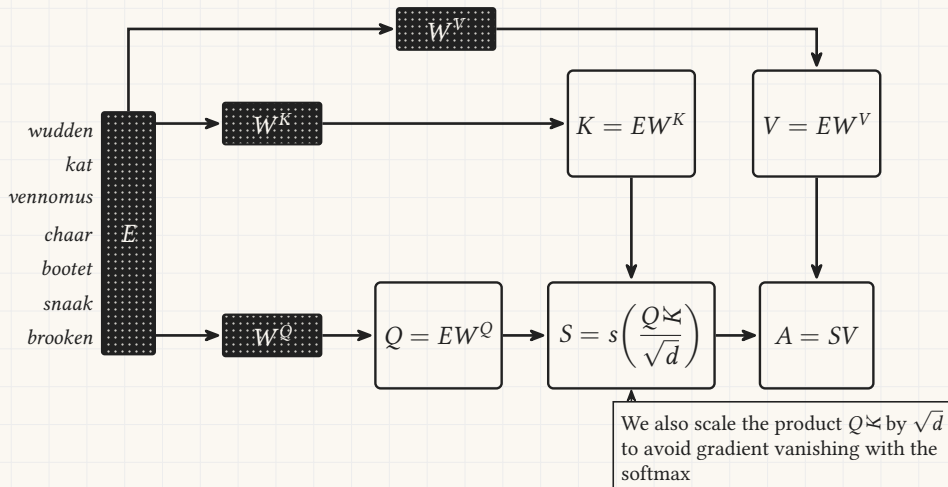
Abstraction of an attention mechanism



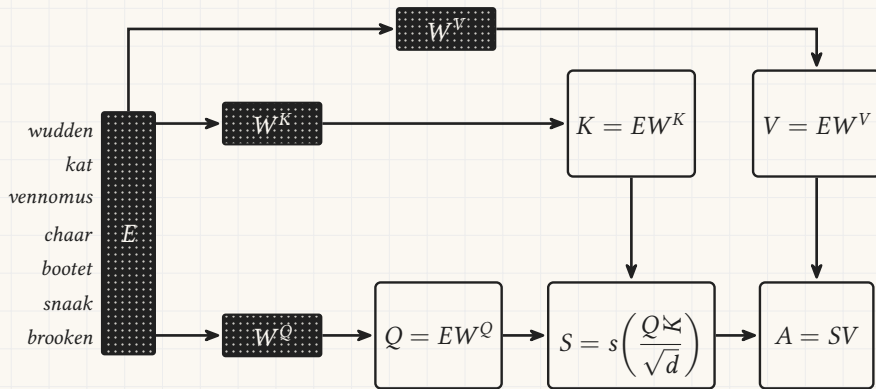
Abstraction of an attention mechanism



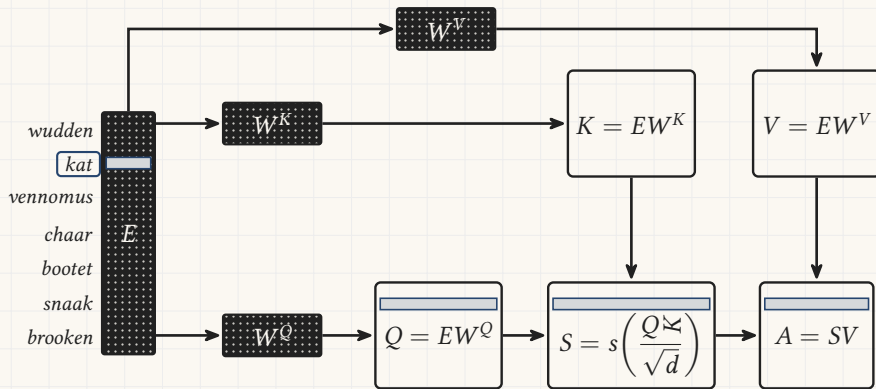
Abstraction of an attention mechanism



Attention from the perspective of information retrieval



Attention from the perspective of information retrieval



Attention from the perspective of information retrieval

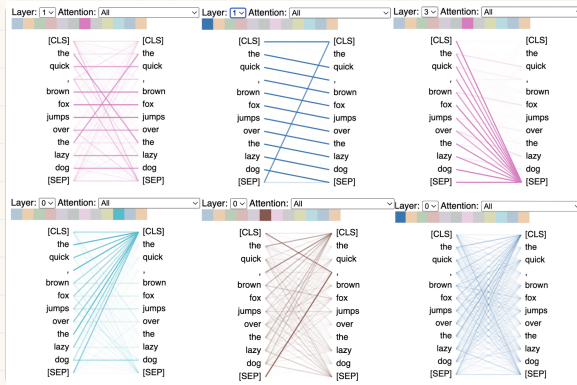
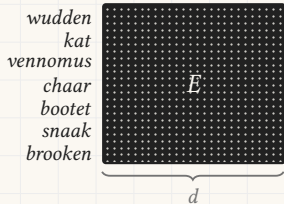
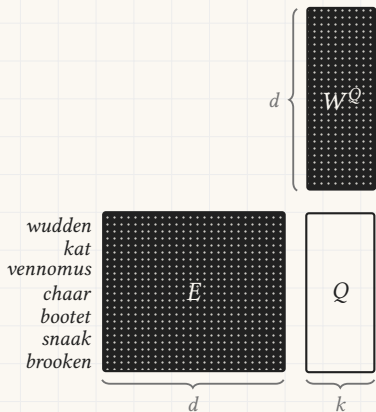


Image from: <https://www.comet.com/site/blog/explainable-ai-for-transformers/>

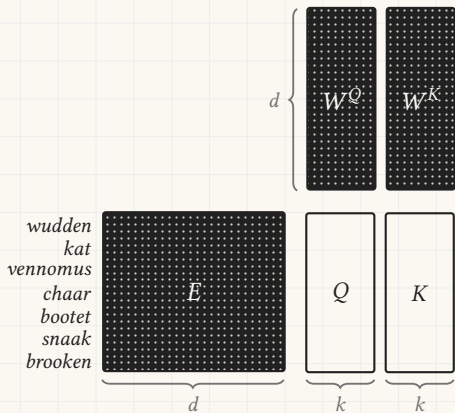
The shapes behind one attention map



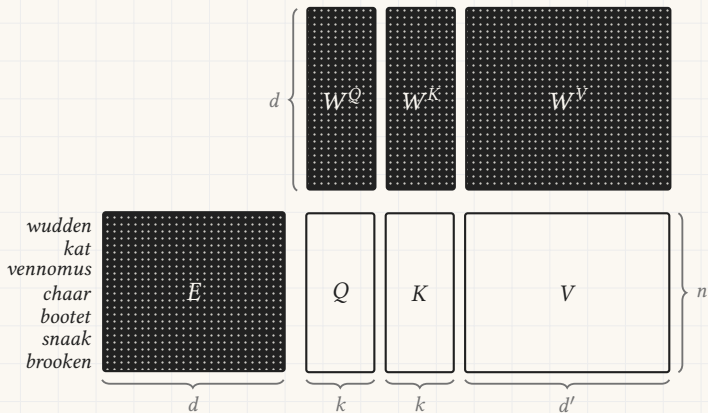
The shapes behind one attention map



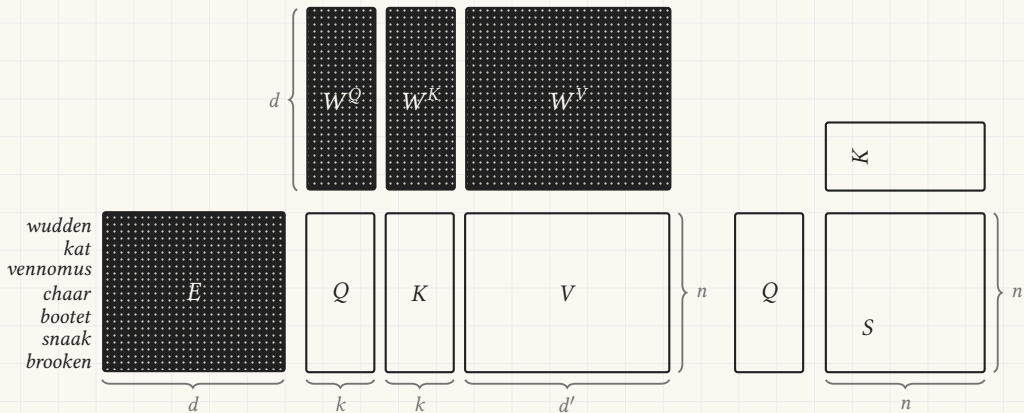
The shapes behind one attention map



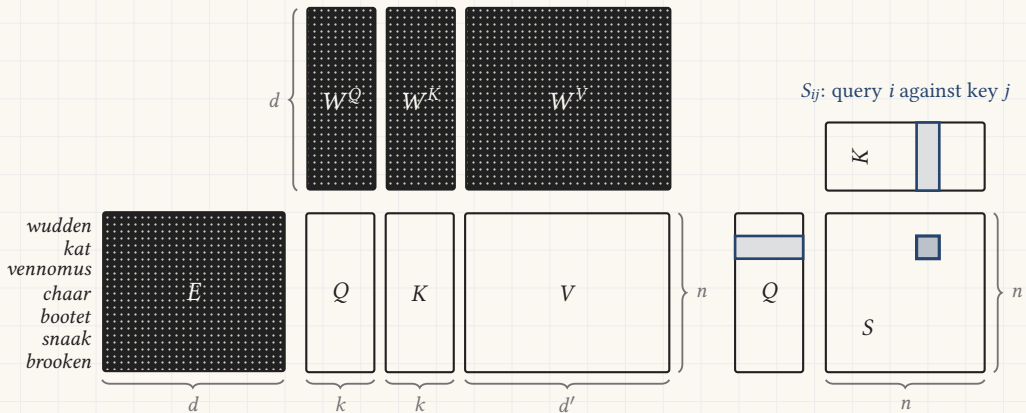
The shapes behind one attention map



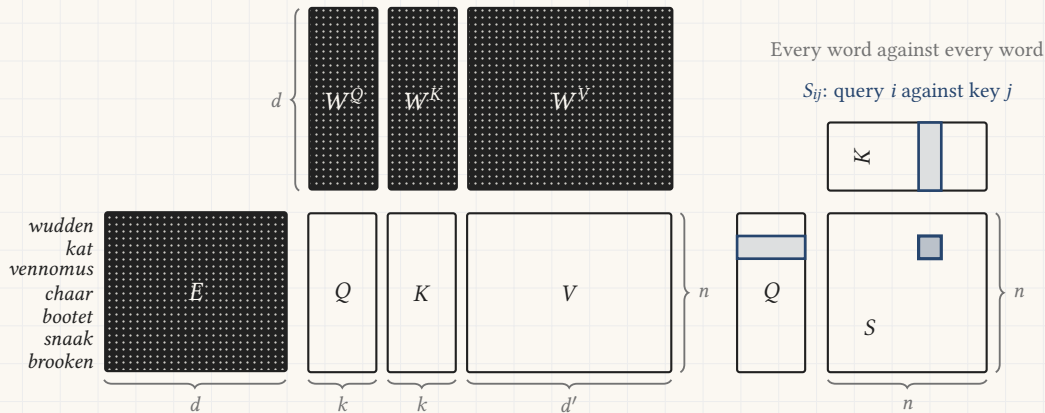
The shapes behind one attention map



The shapes behind one attention map



The shapes behind one attention map



4

Section 4

Multi-headed attention

Multi-headed attention

Remember that we need multiple attention maps.

Alice tried golf and tennis and she liked them

Multi-headed attention

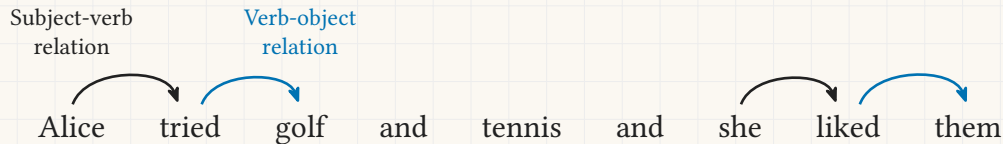
Remember that we need multiple attention maps.

Subject-verb
relation



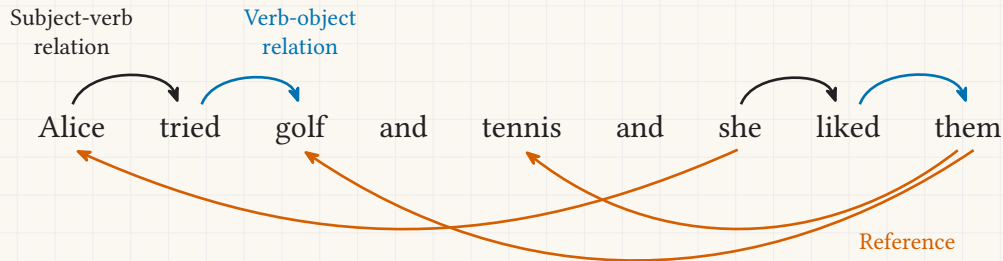
Multi-headed attention

Remember that we need multiple attention maps.



Multi-headed attention

Remember that we need multiple attention maps.

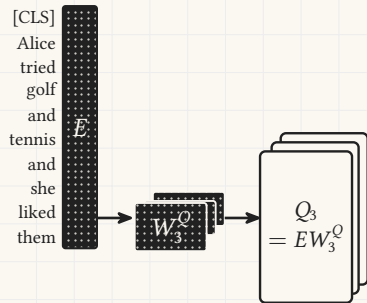


Multi-headed attention mechanism

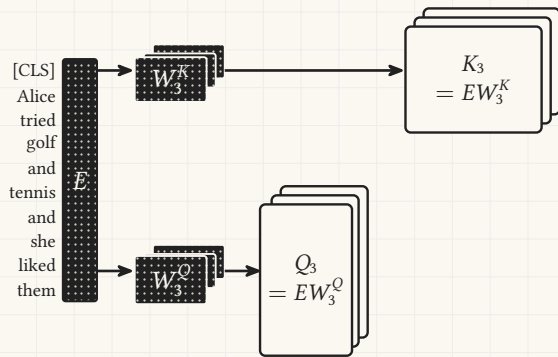
[CLS]
Alice
tried
golf
and
tennis
and
she
liked
them



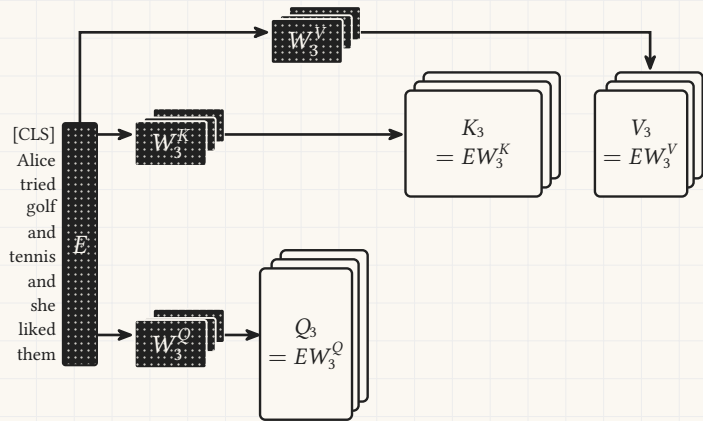
Multi-headed attention mechanism



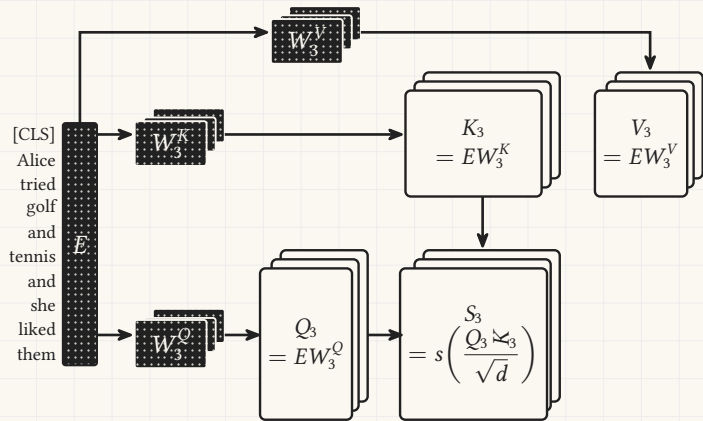
Multi-headed attention mechanism



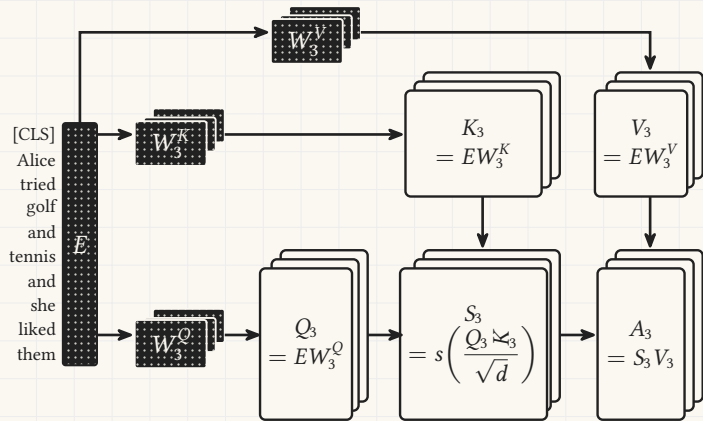
Multi-headed attention mechanism



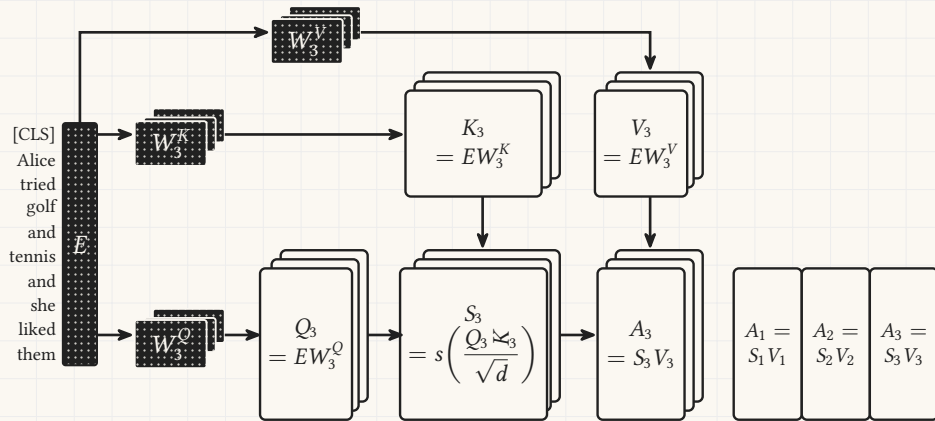
Multi-headed attention mechanism



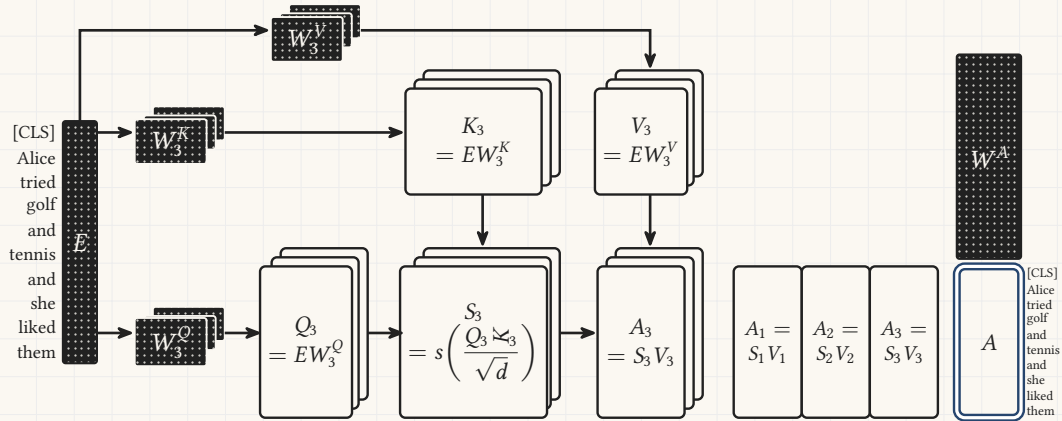
Multi-headed attention mechanism



Multi-headed attention mechanism



Multi-headed attention mechanism



5

Section 5

Cross-attention

Cross-attention

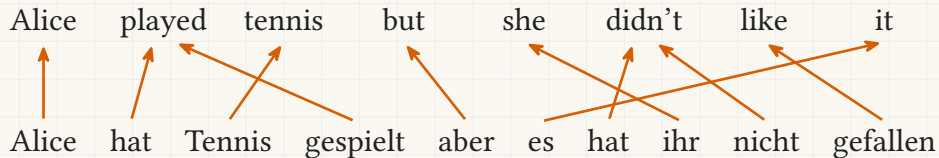
- Consider the problem of translating from English to German.

Alice played tennis but she didn't like it

Alice hat Tennis gespielt aber es hat ihr nicht gefallen

Cross-attention

- Consider the problem of translating from English to German.
- A different attention mechanism is needed.



Abstraction of a cross-attention mechanism

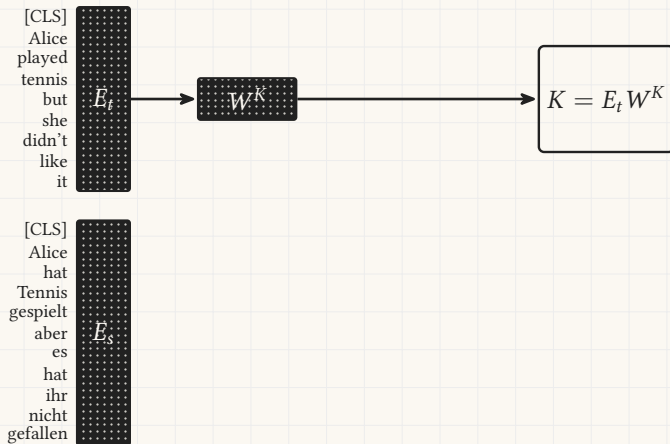
[CLS]
Alice
played
tennis
but
she
didn't
like
it

E_t

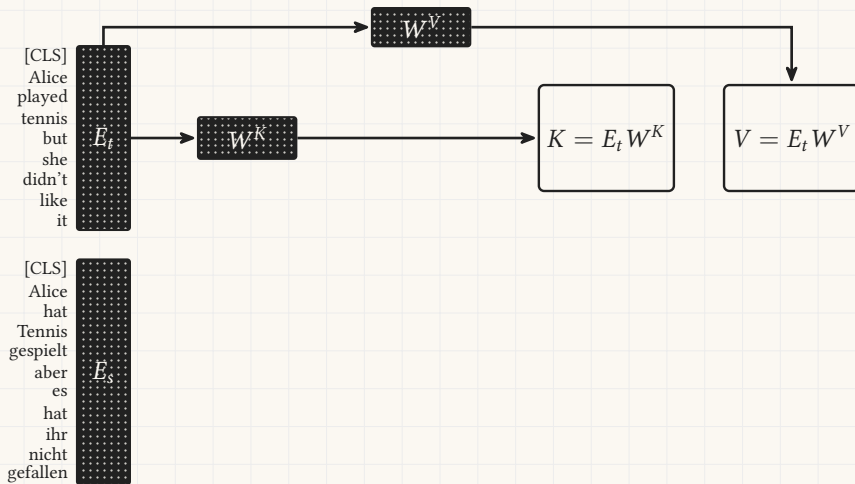
[CLS]
Alice
hat
Tennis
gespielt
aber
es
hat
ihr
nicht
gefallen

E_s

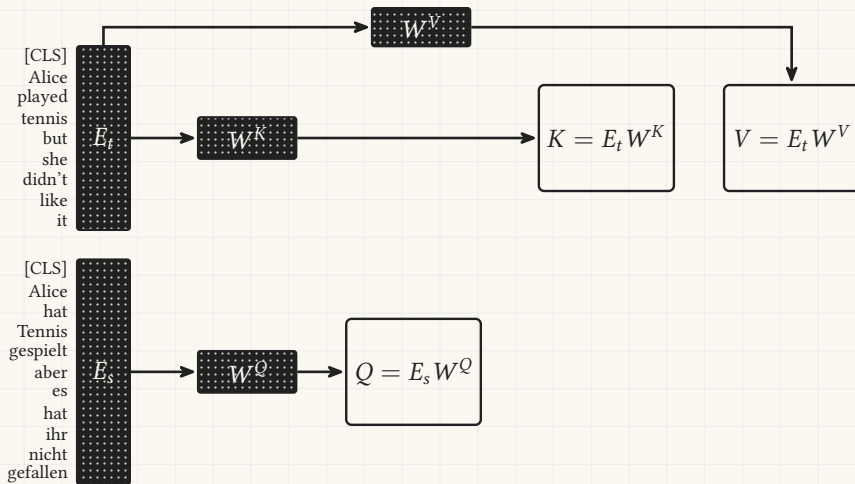
Abstraction of a cross-attention mechanism



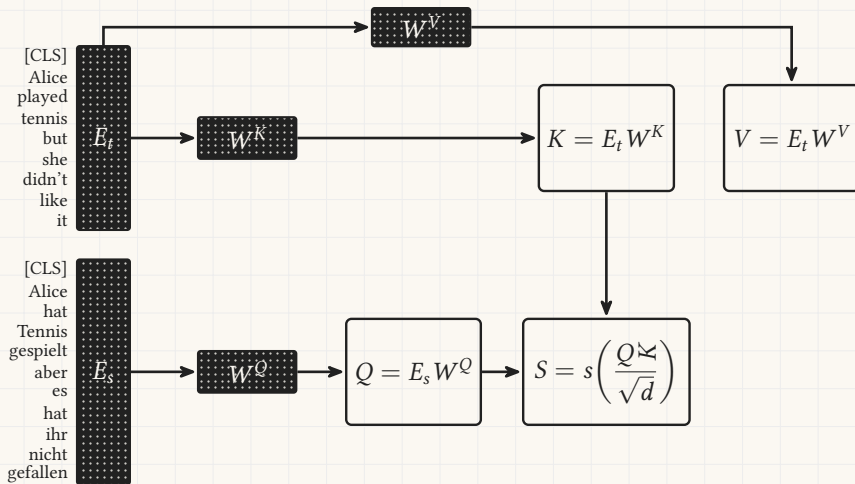
Abstraction of a cross-attention mechanism



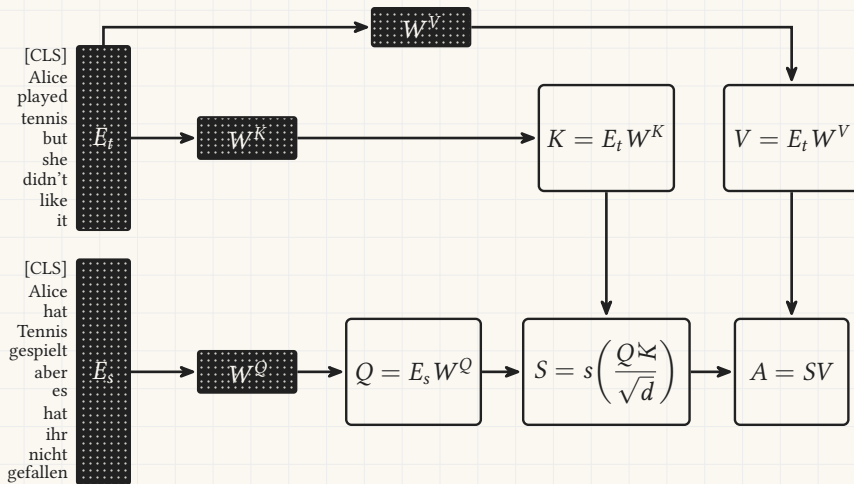
Abstraction of a cross-attention mechanism



Abstraction of a cross-attention mechanism



Abstraction of a cross-attention mechanism



6

Section 6

Masked self-attention

Masked self-attention

- Consider the task of generating text for answering questions.

Why did the bicycle fall over

Masked self-attention

- Consider the task of generating text for answering questions.

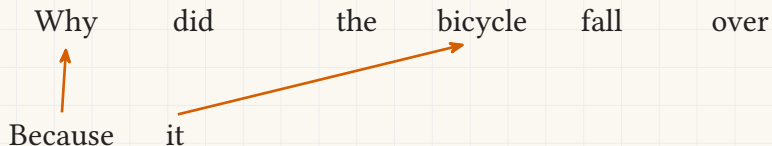
Why did the bicycle fall over

↑

Because

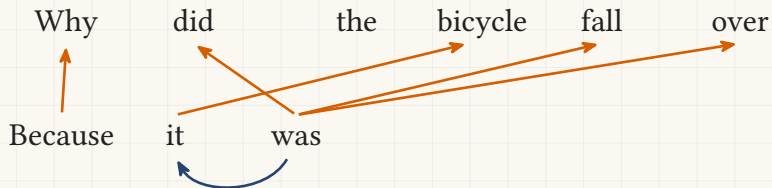
Masked self-attention

- Consider the task of generating text for answering questions.



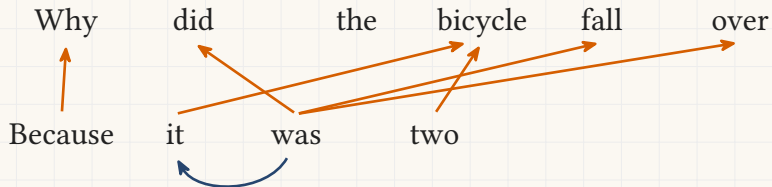
Masked self-attention

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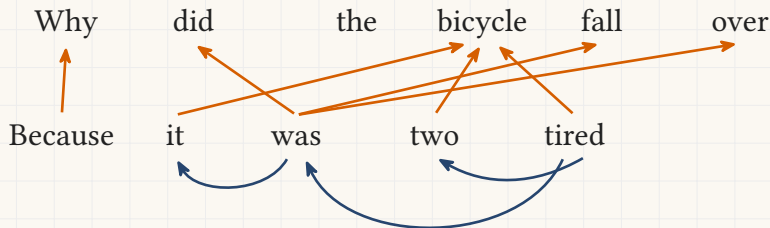
Masked self-attention

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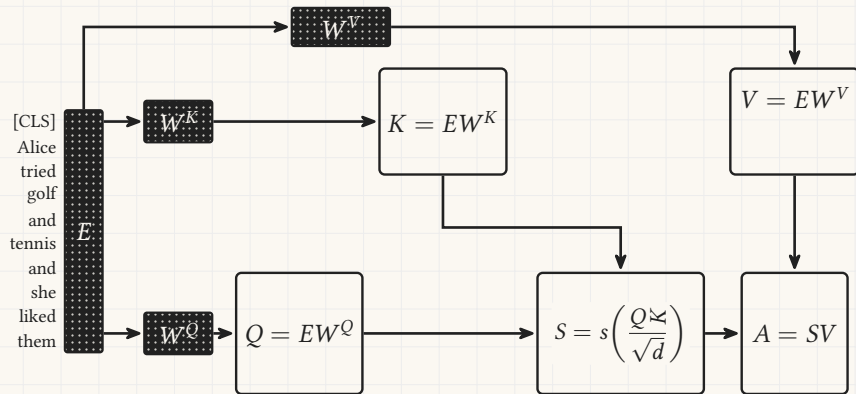
Masked self-attention

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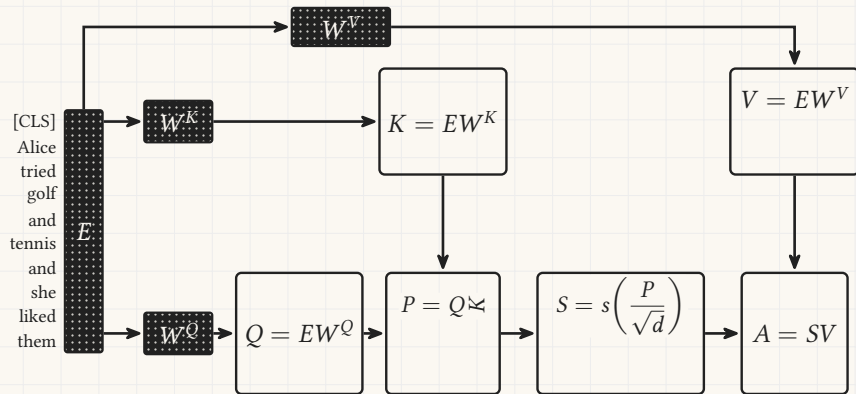


For generating sentences, we need an attention mechanism that only looks at what has been produced so far.

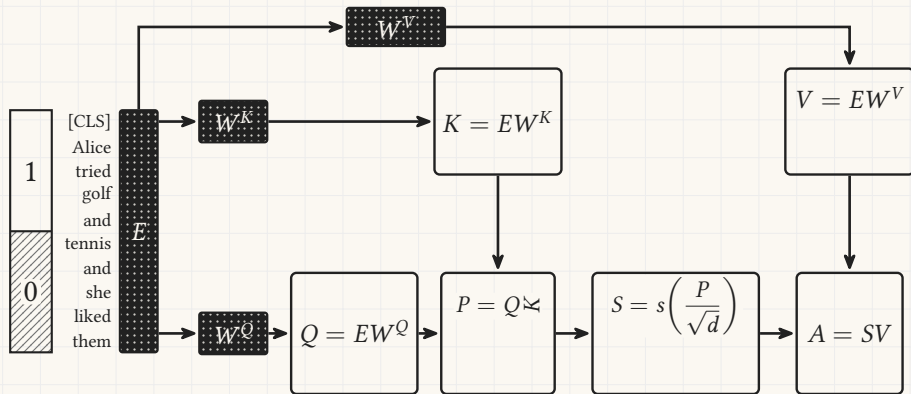
Masked self-attention



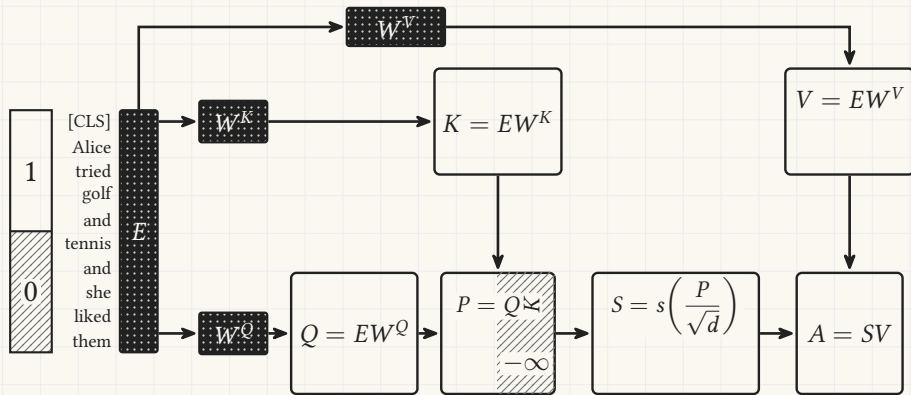
Masked self-attention



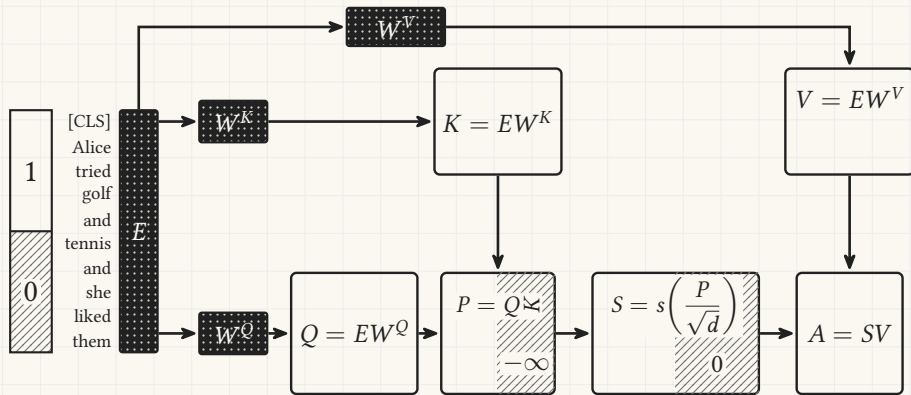
Masked self-attention



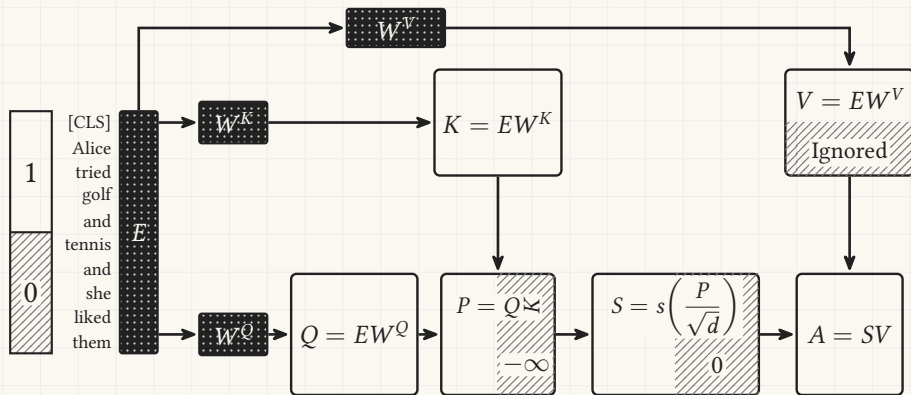
Masked self-attention



Masked self-attention

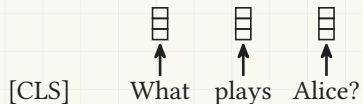


Masked self-attention

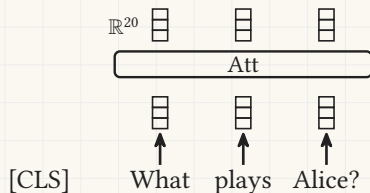


Note how the final embeddings depend only on the embeddings of the unmasked tokens!

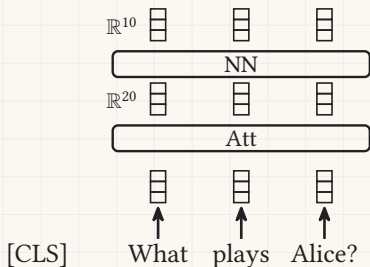
A simplified transformer architecture



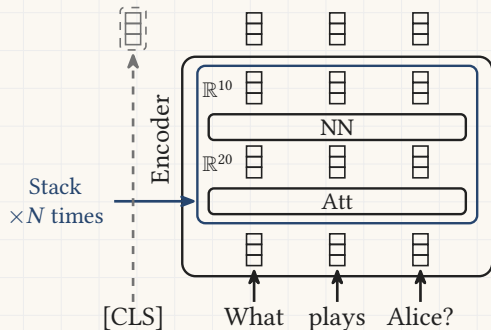
A simplified transformer architecture



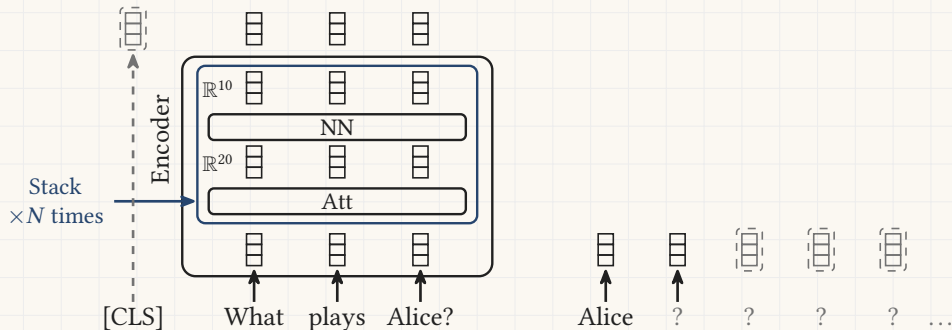
A simplified transformer architecture



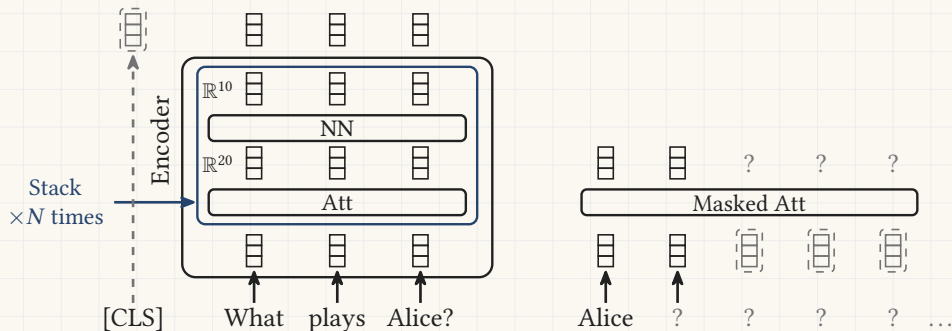
A simplified transformer architecture



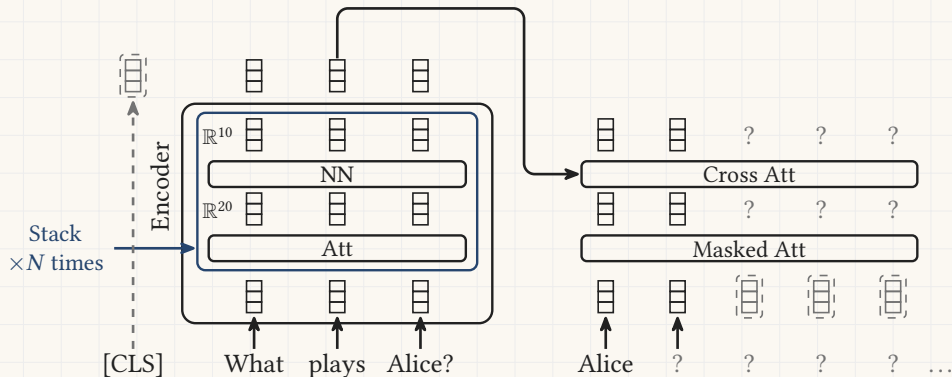
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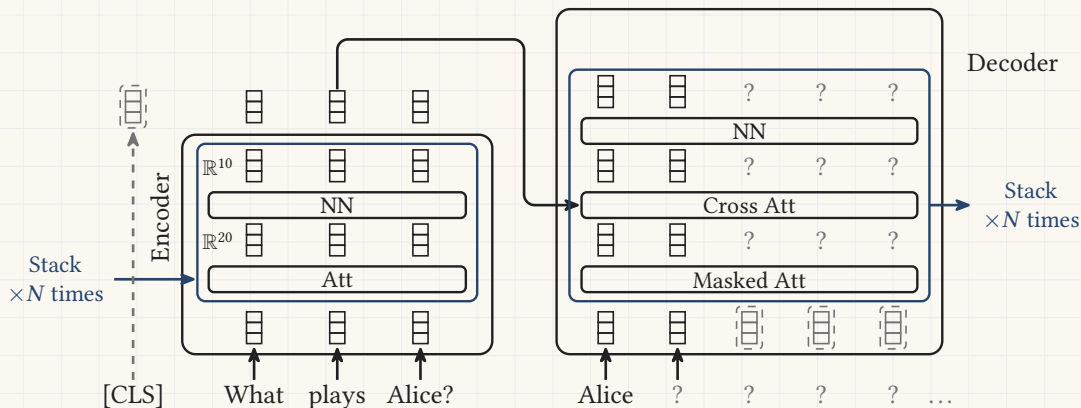
A simplified transformer architecture



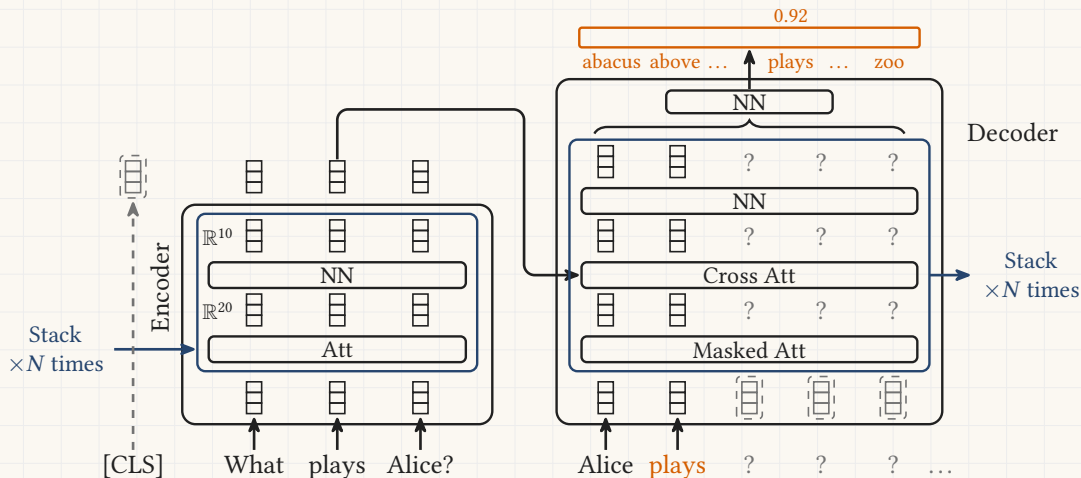
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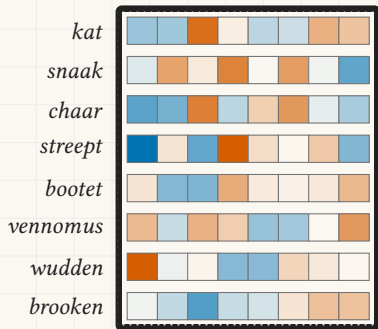
7

Section 7

Training a transformer

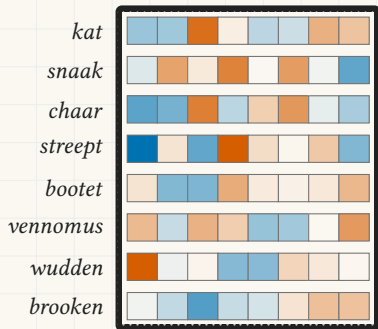
The initial embeddings

One row of numbers per word



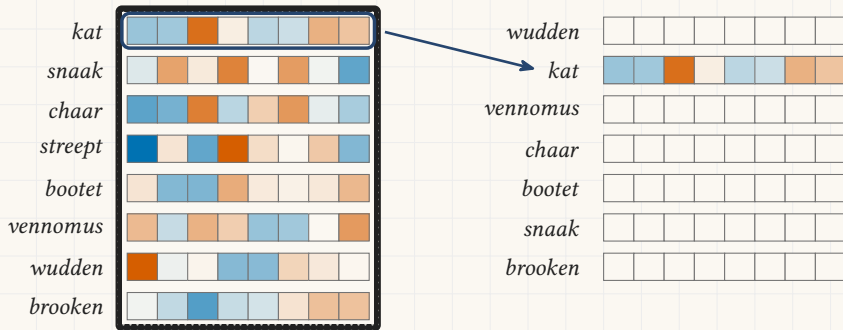
The initial embeddings

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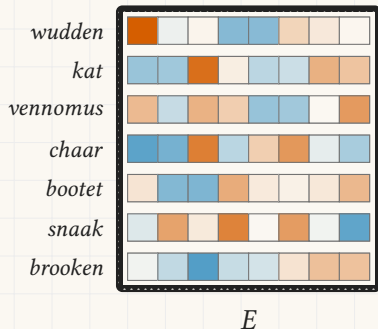
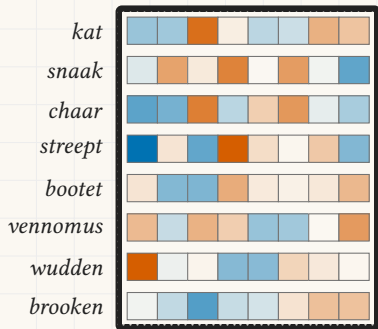
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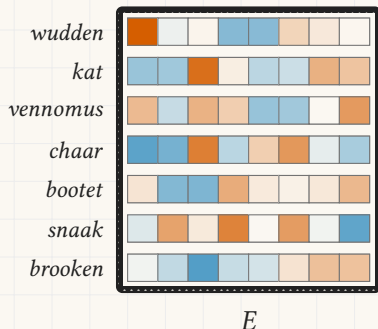
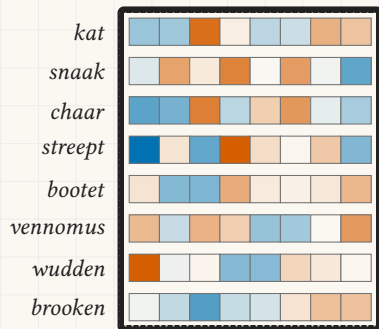
The initial embeddings

One row of numbers per word



The initial embeddings

One row of numbers per word



These numbers are weights too, like W^Q , W^K and W^V . Training guesses them

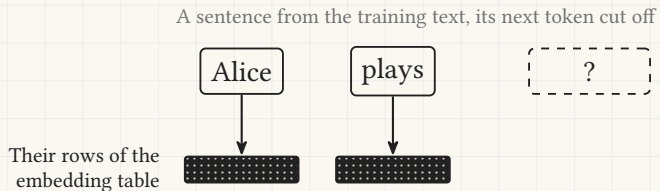
E

The embeddings are guessed in training

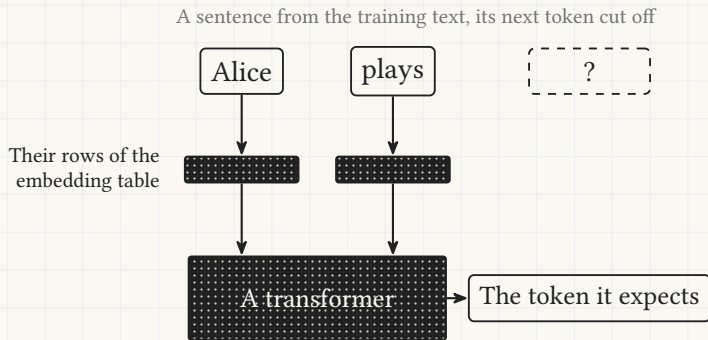
A sentence from the training text, its next token cut off

Alice plays ?

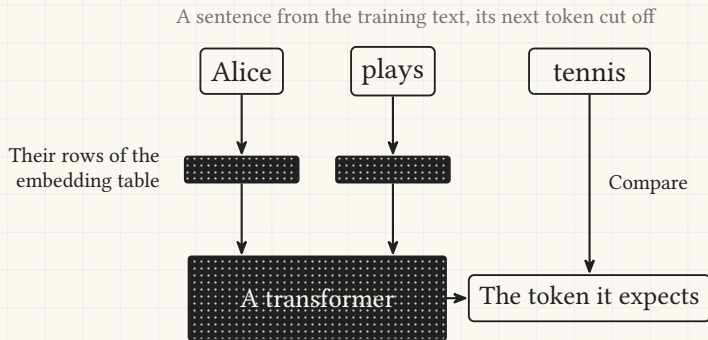
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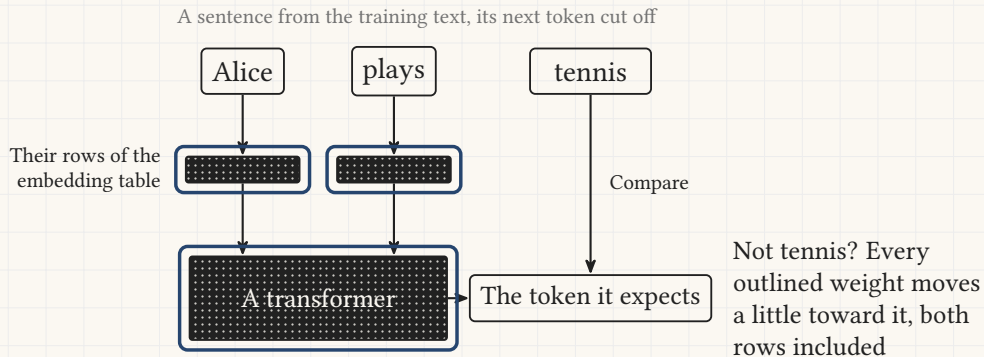
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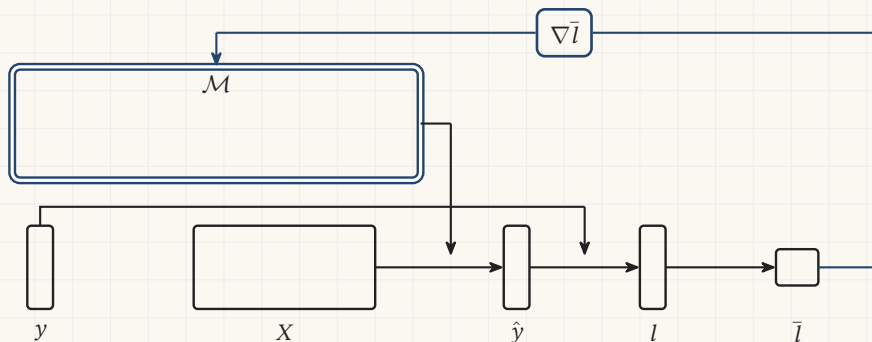
The embeddings are guessed in training



Training a transformer

A row is text with its next token cut off. Every position of every document gives one.

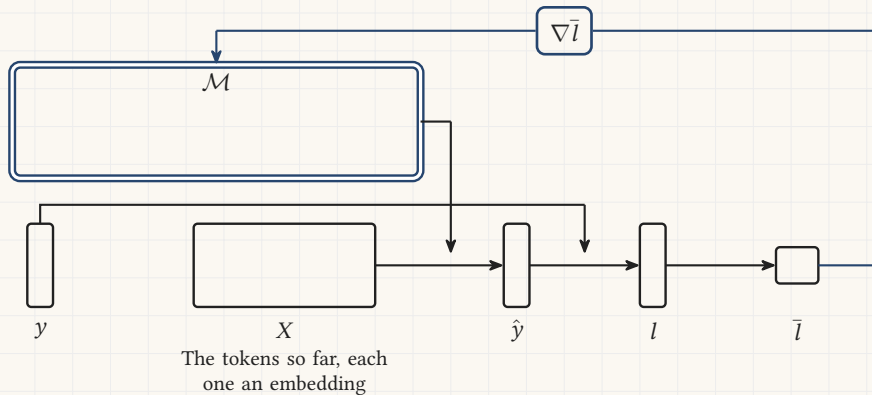
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



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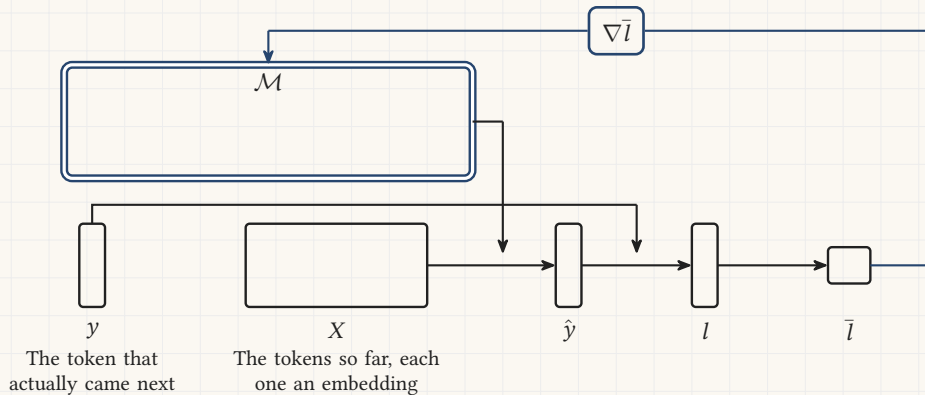
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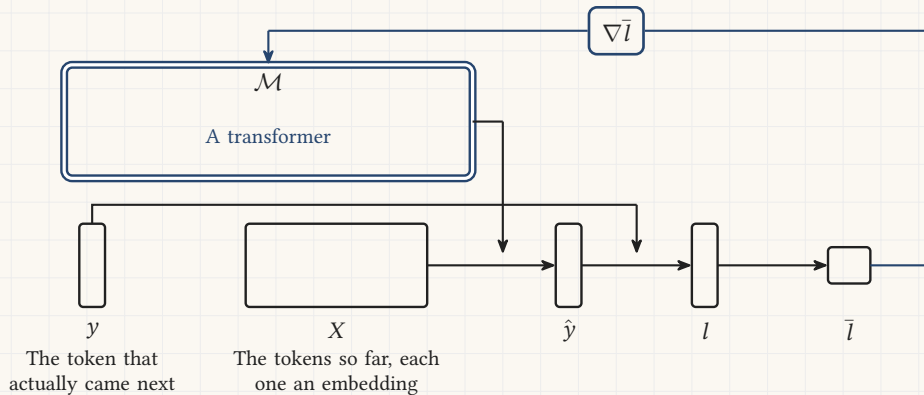
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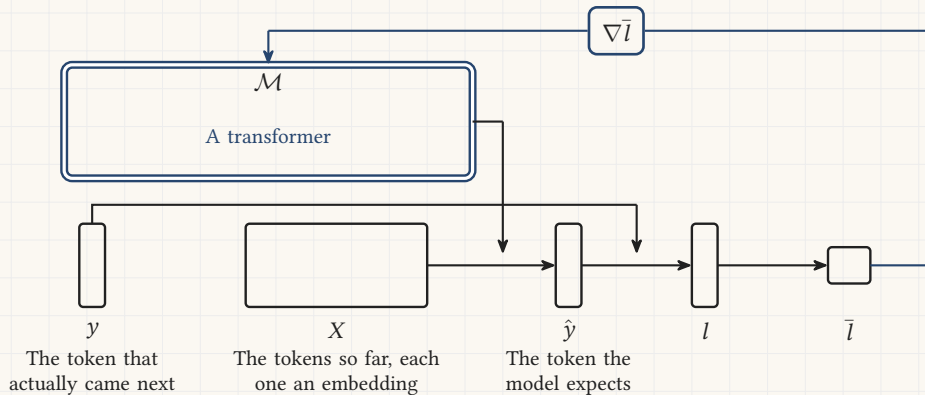
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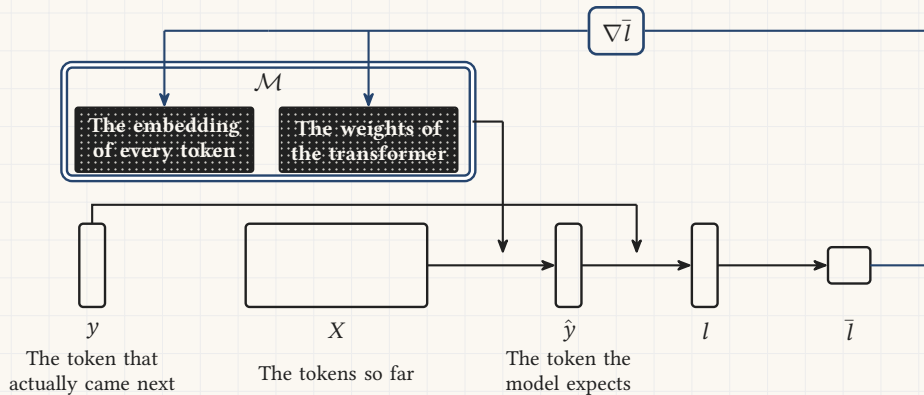
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The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



One row of probabilities per position

Tokens so far

Once

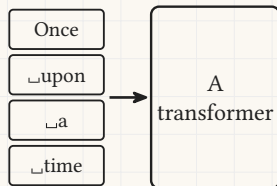
␣upon

␣a

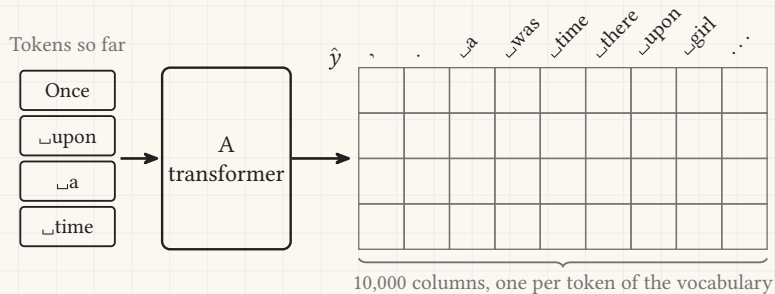
␣time

One row of probabilities per position

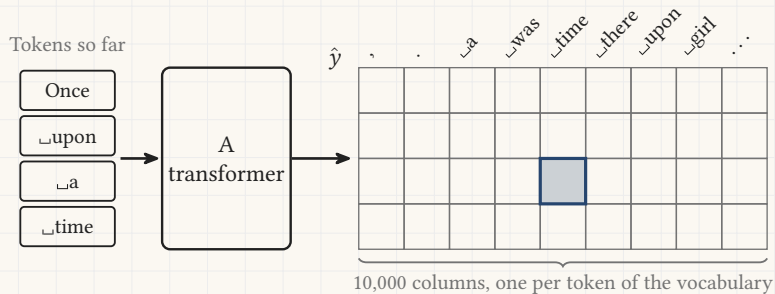
Tokens so far



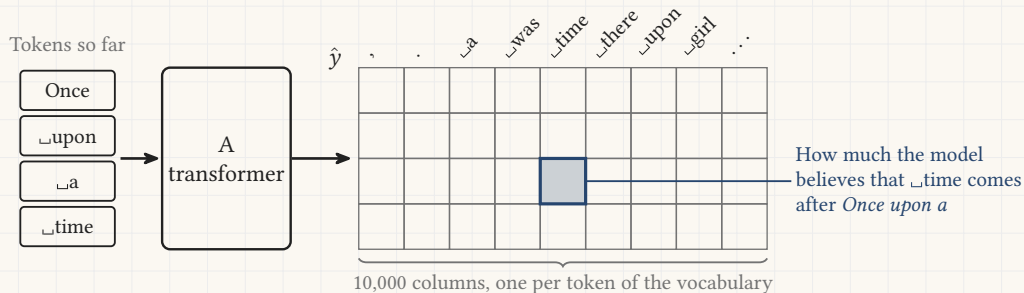
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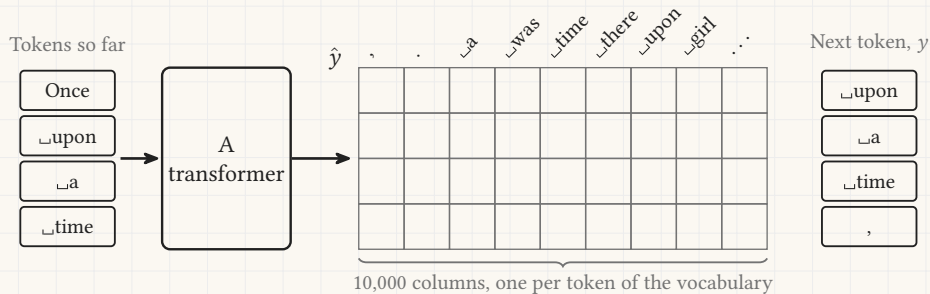
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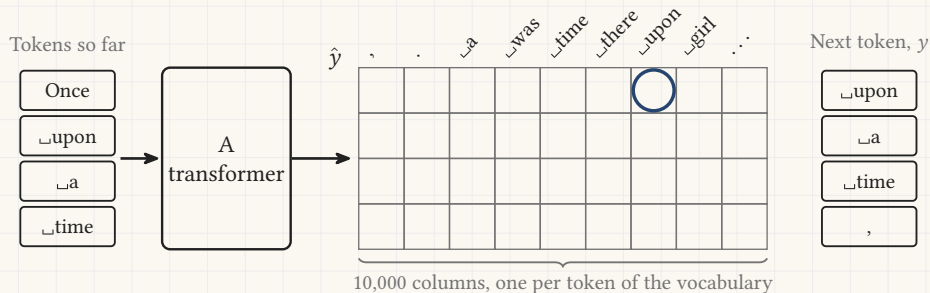
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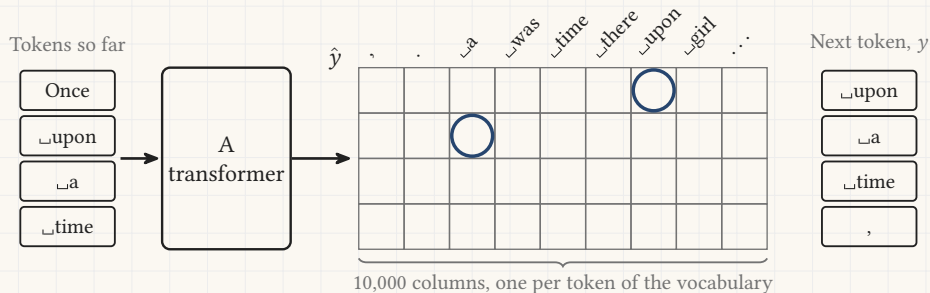
The loss reads one probability per row



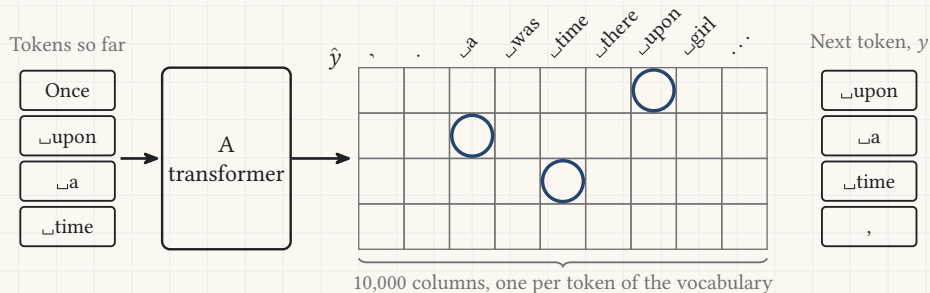
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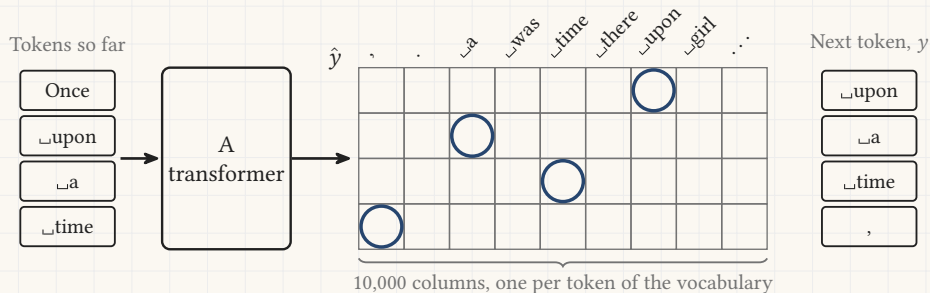
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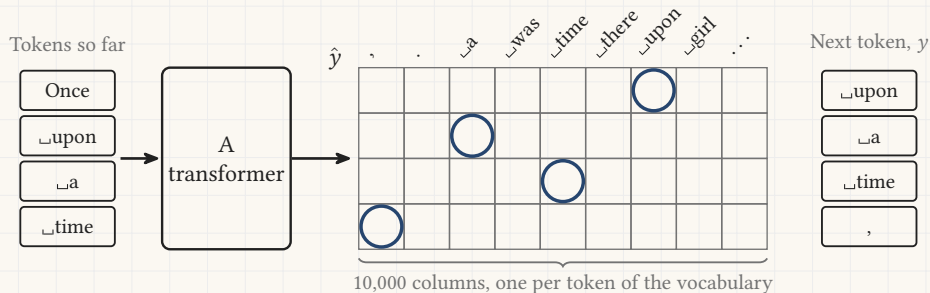
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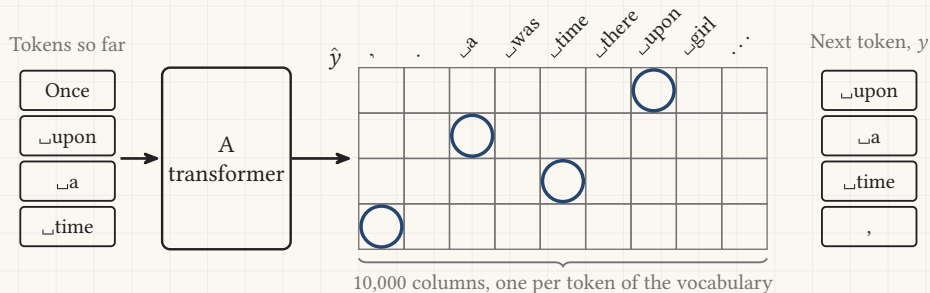


The loss reads one probability per row



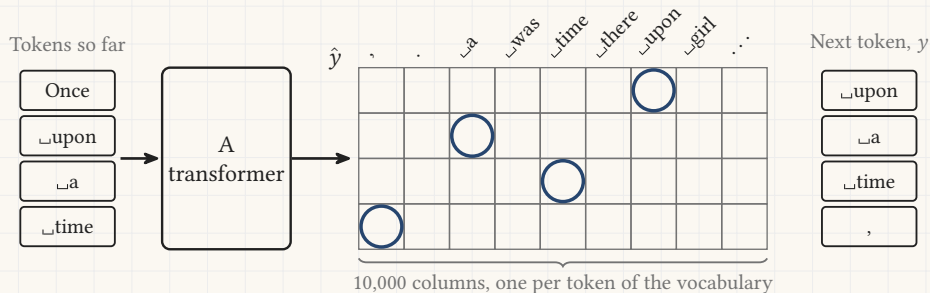
$$\bar{l} = -\frac{1}{n} (\log \hat{y}_{1, \text{upon}})$$

The loss reads one probability per row



$$\bar{l} = -\frac{1}{n} (\log \hat{y}_{1,\text{upon}} + \log \hat{y}_{2,a})$$

The loss reads one probability per row

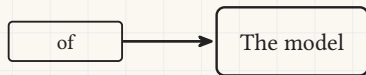


$$\bar{l} = -\frac{1}{n} (\log \hat{y}_{1,\text{upon}} + \log \hat{y}_{2,\text{a}} + \log \hat{y}_{3,\text{time}} + \dots)$$

After training: one token at a time

of

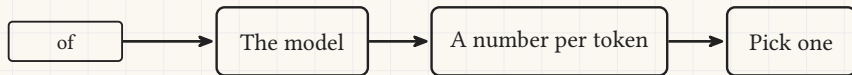
After training: one token at a time



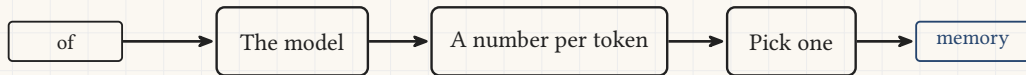
After training: one token at a time



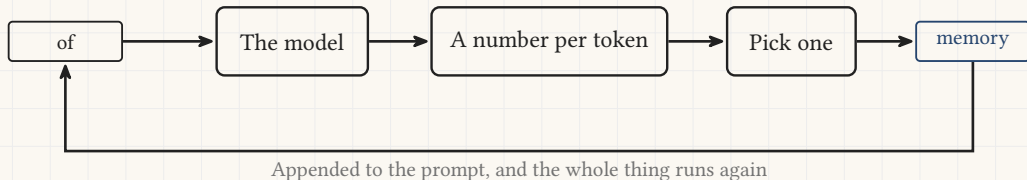
After training: one token at a time



After training: one token at a time



After training: one token at a time



8

Section 8

Positional encodings

Positional encodings

- Consider the sentences *the dog chased the cat* and *the cat chased the dog*.

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- They are permutations of each other, so the attention maps will also just be permutations of each other.

Positional encodings

- Consider the sentences *the dog chased the cat* and *the cat chased the dog*.
- They are permutations of each other, so the attention maps will also just be permutations of each other.
- Note that attention mechanisms do not pay attention to the position of a token in the sentence.

Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
it	0	1	0	0	0	0
was	1	1	0	0	0	0
two	0	0	1	0	0	0
tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

BP

Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
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tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

BP

$$\begin{aligned}BP_{ij} &= (i \div 2^j) \bmod 2 \\ &= \lfloor 2^{-j} i \rfloor \bmod 2\end{aligned}$$

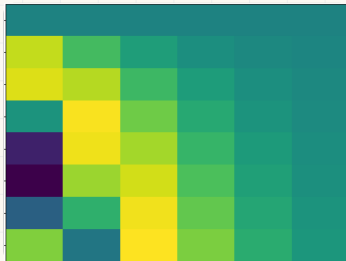
- We would like to create an embedding that encodes the position of the word in the sentence.
- A natural idea is to use a binary encoding of the position.
- Note that the entry i, j of this positional matrix has the form $f(\alpha^{-j}i)$, where $\alpha > 0$ is a scaling factor and f is a periodic function.
- In the binary positional encoding case, $f(x) = \lfloor x \rfloor \bmod 2$.

Binary positional encoding

[CLS]	0	0	0	0	0	0
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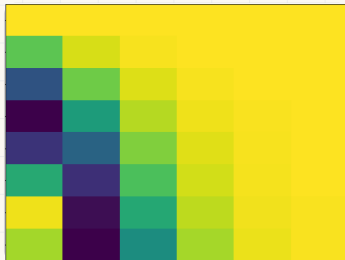
- We generalize this by using sin or cos instead of $f(x) = \lfloor x \rfloor \bmod 2$.

Binary positional encoding

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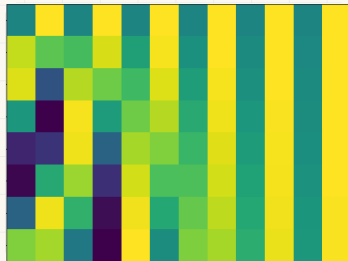
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BP

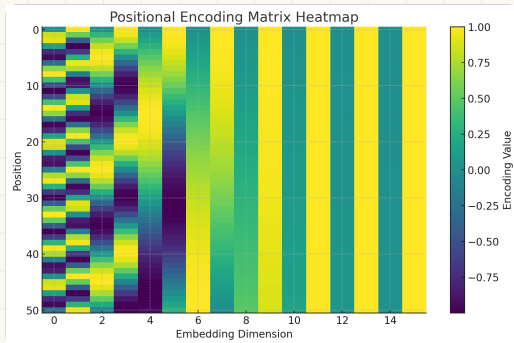
$$\begin{aligned}BP_{ij} &= (i \div 2^j) \bmod 2 \\ &= \lfloor 2^{-j} i \rfloor \bmod 2\end{aligned}$$



- We actually use both sin and cos and then interleave them.

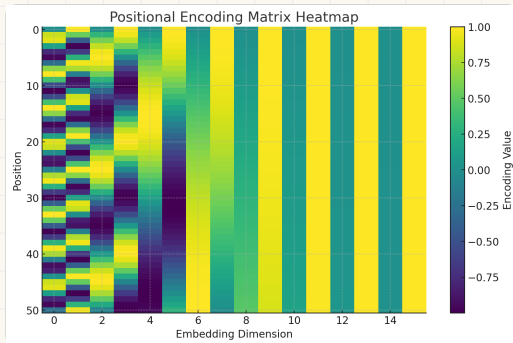
Positional encodings

- The positional encoding matrix p has T rows and d columns.



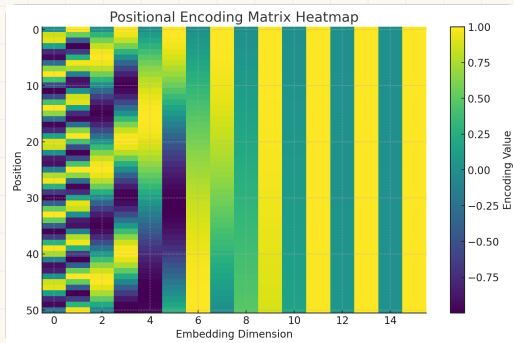
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- The value d is the dimensionality of the positional embedding.



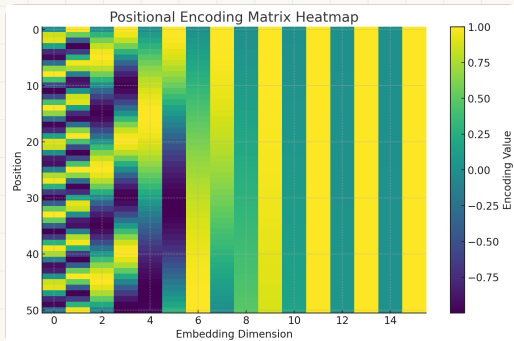
Positional encodings

- The positional encoding matrix p has T rows and d columns.
- The value T is the maximum number of tokens in a sentence.
- The value d is the dimensionality of the positional embedding.
- The encoding of token i is defined through the formulas

$$p_{i,2j} = \sin(\alpha^{-2j} i) \text{ and}$$

$$p_{i,2j+1} = \cos(\alpha^{-2j} i),$$

$$\text{for } j \leq d/2 \text{ and } \alpha = 10^{4/d}.$$



Why adding and not concatenating?

- Concatenation substantially increases the dimensionality of the embedding, increasing the number of parameters to be trained.

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- With sufficient dimensionality, since the positional embeddings are somewhat statistically independent from the token embeddings, both embeddings are somewhat orthogonal. Hence, the addition still preserves information about the two vectors.
- Since most of the operations done in an attention mechanism are linear, the final embedding is somewhat the addition of the result of the operations applied to both the positional and the token embeddings.

9

Section 9

Residual blocks

The degradation problem

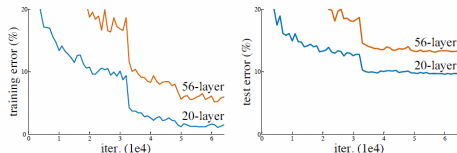


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

- When deeper networks are able to start converging, a *degradation problem* appeared. Training performance degrades rapidly. This is not because of overfitting.

Figure 1 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

The degradation problem

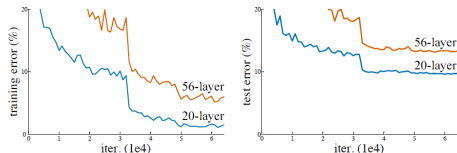


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

Figure 1 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

- When deeper networks are able to start converging, a *degradation problem* appeared. Training performance degrades rapidly. This is not because of overfitting.
- The problem is mostly due to difficulties of information propagating, both forward and backward, through the layers in the network.

Residual blocks

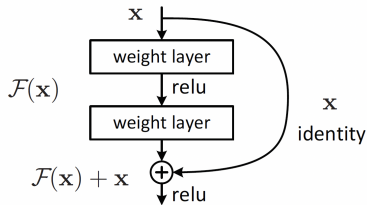


Figure 2. Residual learning: a building block.

Figure 2 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

Residual blocks

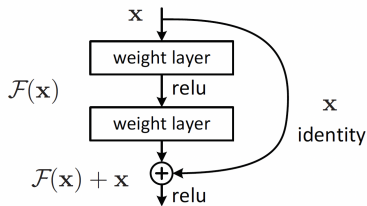


Figure 2. Residual learning: a building block.

- To address this problem, residual blocks were introduced. They work by adding the output of a layer to the output of a few layers ahead.

Figure 2 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

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Section 10

Transformer architecture

Transformer architecture

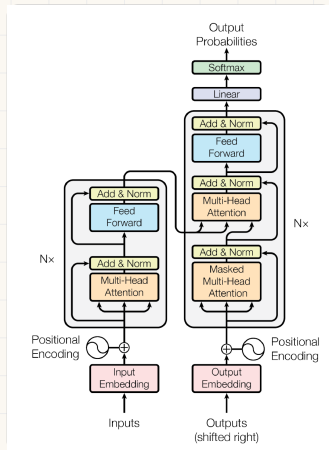


Figure 1 of

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser and Illia Polosukhin.

Attention Is All You Need.

Advances in Neural Information Processing Systems 30 (NeurIPS 2017).

<https://arxiv.org/abs/1706.03762>